

Recognition memory decisions made with short- and long-term retrieval¹

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Abstract

The present research explores the way retrieval from both short- and long-term memory is used to make recognition decisions about the contents of short-term memory. Pictures are presented in a short list followed by a test picture, either a target or foil. The research contrasts four paradigms: VM: target and foil responses to a given stimulus change from trial to trial; CM: the responses do not change from trial to trial; AN: every trial uses new stimuli; MIXED: combinations of VM, CN, and AN occur on each trial. These conditions vary the way long-term memory (traces from prior trials) affect current trial recognition decisions. A new paradigm is used in which a given picture is equally often presented and tested in both VM and CM. A model is presented that shows the way short- and long-term retrieval work together to produce the observed accuracy and response time performance. The model is used to predict published data from a traditional paradigm (Nosofsky et al., 2021) and the new data. It predicts quite well the accuracy and response time results from 376 conditions, while holding almost all parameter values constant across groups and conditions, lending support to its assumed processes.

198 words

¹ A version of this article is available at <https://psyarxiv.com/h8msp> . This version contains appendices giving additional information and the code used to produce the predictions of the model.

Recognition memory decisions made with short- and long-term retrieval

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This report shows how the varied processes of storage in and retrieval from both short- and long-term memory work together to allow memory to operate. We do this by modeling data from short-term probe recognition experiments. These studies require a decision whether a test item had been studied on a just presented list ('target'), or had not ('foil'). The items on the lists are presented sequentially, and are few in number (varying from 2 to 9 in the three studies analyzed in this report). This report analyses and models 13 groups of participants from three studies, containing a total of 376 conditions, each providing a measure of accuracy and mean response time. The success of the model in accounting for the results gives confidence that it captures the major processes of memory and the way they work together. The processes of the model have separately received support in many prior publications, but not previously together with common parameter values for such a wide range of conditions.

It has long been recognized that most memory tasks require retrieval from both short- and long-term memory (e.g. Atkinson & Shiffrin, 1968, for recall; Mandler, Pearlstone, and Koopmans, 1969, for recall and recognition; Atkinson and Juola, 1974, for recognition). For recognition testing these reports also give evidence for the use of two processes, recall (defined as use of the contents of memory traces) and familiarity (defined as summed activation of traces). In this tradition the present modeling makes use of similar assumptions.

Much of the modeling of memory since the 1950s uses accuracy of retrieval as a primary measure. Mostly for modeling recognition, researchers have produced models that predict both accuracy and response time (e.g. Atkinson and Juola, 1974; Ratcliff, 1978). Response times are often used as a measure for recognition paradigms because response times for binary responses are relatively easy to interpret. Especially in recent years models predicting both accuracy and response time have been based on 'evidence accumulation': Probabilistic evidence is accumulated in steps, until sufficient evidence favors one response over the other.

The recognition studies we address present a short list of pictures followed by a test of either a picture from that list (*target*, or *OLD*), or a picture not from that list (*foil*, or *NEW*). To model recognition memory we assume that two processes contribute to performance: *recall* when the design allows learning to occur, and *familiarity*. As is typical, we assume that the test item and context are used to probe memory and activate traces to a degree determined by the strength of the trace and its similarity to the test item. Familiarity has two components: summed activation of short-term traces and summed activation of long-term traces. This is necessary for modeling short-term recognition because much research including the present studies shows that long-term traces contribute to performance even though the task seems to require only short-term activation. Some of this evidence comes from studies in which targets on one list are foils on other lists, and vice versa (called 'varied mapping' or VM; Schneider & Shiffrin, 1977; Shiffrin & Schneider, 1977)). If only traces of items from the current list determine performance then

VM would perform no worse than AN, in which every trial uses new pictures. However VM performance is much inferior, because long term traces are activated and the responses they encode are random with respect to the current trial. To improve VM performance it is likely that participants attempt to use current context to focus retrieval as much as possible on short-term traces, but the data show that such focusing is imperfect.

It is the aim of this article to produce a general model for the way that short-term recognition memory involves and uses retrieval from both short- and long-term memory. To pursue this goal, we build upon a good deal of research by Nosofsky and colleagues that have explored short-term recognition, measuring and modeling both accuracy and response time (Nosofsky, Little, Donkin, & Fific, 2011; Donkin & Nosofsky, 2012a, b; Nosofsky, Cao, Cox, & Shiffrin, 2014a,b; Nosofsky, 2011; Nosofsky, 2016; Nosofsky & Donkin, 2016; Cao, Nosofsky, & Shiffrin, 2017; Cao, Shiffrin, & Nosofsky, 2018). Some of these studies, including the most recent by Nosofsky, Cao, Harding, Cousineau, and Shiffrin (2021), contrasted consistent mapping (CM) of targets and foils across trials with varied mapping (VM) of targets and foils across trials. In both CM and VM the targets and foils are selected from a small set of total stimuli. In CM tasks one set of items are always targets and another set of items are always foils, as trials continue. CM designs allow learning to take place, learning what response to give for each test item. Learning of course improves recognition decisions. VM tasks randomly assign targets or foils as trials continue. Activation of long-term traces, those from prior trials, are random with respect to the correct decision on the current trials, so their activation harms performance, in ways we will lay out. The Nosofsky et al. (2021) study and the present studies also contained AN conditions (every trial used new items). AN conditions are a control in which activation of long-term traces is minimal. Both studies also used ‘mixed’ conditions (each trial contained 2 or 3 of the VM, CM, and AN stimuli). Mixing makes it likely the same strategies and processes are used for the three kinds of stimuli.

There are several related hypotheses that could explain the superiority of CM to VM: 1) In CM but not VM item-response pairings might be stored in the long-term traces. This hypothesis seems unlikely to hold in mixed conditions on which the CM and VM items occur in the same trials. 2) In both CM and VM item-response pairings might be stored in long-term traces, but in CM those responses are used to guide responding (due to the use of learning and recall), but in VM only total activation and not the response mapping is used to produce responding (only familiarity guides responding); the present modeling uses this approach. 3) In both CM and VM item-response pairings might be stored and retrieved, causing harm to VM because the long-term retrievals in VM are half the time incorrect for the current trial (as assumed in Nosofsky et al., 2021).

Modeling details aside, the design of traditional short-term recognition studies makes it possible that CM could be superior to VM even without learning. In CM conditions a given target item is studied on many trials but tested only on some of those trials. However, a given foil item only occurs when tested, and the number of such tests is the same for CM and VM. Thus the extra target presentations in CM could make targets more ‘familiar’ than foils. This long-term familiarity difference is not present in VM due to the random pairing of responses over trials. According to some models the greater long-term familiarity for targets than foils could make CM superior to VM. The present studies eliminate this possibility, as described next.

METHOD

Overview:

The present studies were designed to equate the retrieval from long-term memory of both targets and foils in both VM and CM conditions, but nonetheless allow the response key for each stimulus in CM to be learned. Our design insures that any superiority of CM over VM would have to be due to learning. The studies use a probe recognition paradigm: A short list of pictures is presented one per second for study. The lists vary in length from 2 to 8 (3-9 in the published study). The test picture is one half the time a target ('OLD') and one half the time a foil ('NEW'). The participant gives a binary response whose accuracy and response time is measured. Some groups received 'pure' lists: All trials were either AN, VM, or CM. Other groups received 'mixed' lists: each trial had pictures of two of these three types (all three types in the published study).

The new design presented the pictures for study on a given trial on either the left side of fixation (left columns in Figure 1) or the right side of fixation (right columns in Figure 1). Examples of four CM trials for a given participant in this new design are given in the four panels of Figure 1. A test picture follows, and when the trial was one on the left side a key press on the left side of the keyboard is used to indicate that a target has been tested and a key press on the right is used to indicate that a foil has been tested. On a right side trial, a key press on the right is used to indicate that a target has been tested and a key press on the left is used to indicate that a foil has been tested. In the CM conditions there are two sets of N pictures, one always used for study lists on the left and the other always used for study lists on the right. In both cases the foil is from the set on the other side. With this design, a given CM picture, whether a target or foil, always requires a response key on the same side, allowing the key-press response to be learned. The VM conditions have 2N pictures; on each trial a random subset is selected for study and assigned randomly to side; in this case the key press cannot be learned.

In this design (and also the standard paradigm), *short-term trace activation* is larger for targets than foils, in both CM and VM, because a tested target strongly matches one trace due to recent study. In the present paradigm long-term trace activation is the same for targets and foils, not only for VM but also for CM. Thus it is only learning (the key to press) that can produce CM superiority to VM: There is no extra long-term familiarity for CM targets.

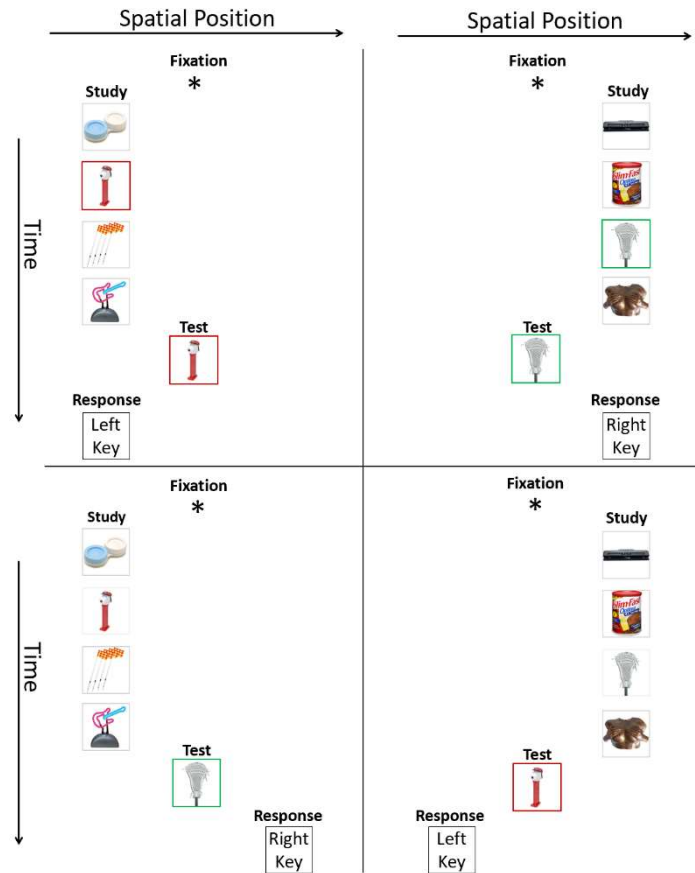


Figure 1: The top panels show left and right side target test trials for set size four that could have occurred in CM, VM, AN, or MIXED conditions. In AN conditions no picture repeats on any trial. In VM conditions a given picture is studied randomly on a given side as trials continue. In CM conditions a picture is always studied on the same side. Thus the bottom two panels show foil tests for CM that could have occurred on other trials: Because the response key for a target is on the side of study, and for a foil is on the opposite side, the comparison of the top and bottom panels show that for CM the same response key is used for the same picture, whether a target or foil.

Figure 1 about here

Method Details

Two very similar experiments were carried out with about a one-year delay between.

Probe Recognition.

On each trial, the participant sees a series of pictures to remember, presented one at a time, each in the same location on the computer monitor. Following this study list a single test picture is presented; with probability 0.5, the test picture will be a randomly chosen picture from the study list (called a 'target', or OLD); with probability 0.5, the test picture was not on the study list (called a 'foil', or NEW). The study lists have 2, 4, or 8 pictures, the number chosen randomly on each trial.

With probability 0.5, the study list and test picture are presented on the left side of screen center, and otherwise on the right side of screen center. There are two keyboard letters used to respond: 'F' (the key on the left) and 'J' (the key on the right). The participant is instructed to respond to a target with the key on the side of the presentation, and to a foil with the other key. Participants are instructed to put index fingers on the two response keys at all times and to respond as fast and as accurately as possible. They are given practice for a brief period at the start of a session. They are not informed of the differences across conditions in stimuli or response mappings (i.e. CM, VM, AN), and are not told the length of the coming list prior to each trial.

Stimuli and Apparatus.

The stimuli are chosen from 2,400 daily life pictures from the website of Talia Konkle (Brady, Konkle, Alvarez, and Olivia, 2008). Each image subtended a visual angle of approximately 7 degrees and is displayed on the designated side of the screen of approximately 20 degrees from screen center and roughly halfway from bottom to top. The experiment is conducted with MATLAB Psychophysics Toolbox (Brainard, 1997) on PCs. All participants are tested in private, sound-attenuated booths.

Trial Timing

On each trial, a fixation point (*) appears on the center of the screen for 0.5 seconds. Each picture in the study list is displayed on the designated side for that trial for 1 sec., with an inter-stimulus-interval between pictures of 0.1 second. Following the study list, for Experiment 1, a test picture appears at screen center; for Experiment 2 a test picture appears at a position half way from the study pictures to screen center (and hence still on the side of presentation). In both experiments the change of position signals to the participant that a test has occurred and a response is needed. The test probe stays on the screen until the key response is registered. A blank screen then lasts for 0.5 second followed by feedback ("correct" or "incorrect") for 1 second. Then the next trial starts.

Groups and conditions:

Across both experiments there are nine groups with different participants. Each group has 9 blocks of 24 trials. The first block is training and not analyzed. The assignment of participants to conditions within group is counterbalanced. For each block half the trials use the left side, half the right side, randomly permuted. The study lists contain either 2, 4, or 8 pictures, these set sizes randomly mixed in each block.

- 1) **Pure VM.** *Used in both Experiments 1 and 2:* Each participant has 16 pictures randomly chosen from the 2400. On each trial the study list pictures are randomly selected from the 16. A foil is randomly selected from those pictures not on the study list.
- 2) **Pure CM.** *Used in both Experiments 1 and 2:* Each participant has 16 pictures randomly chosen from the 2400, 8 assigned to be targets on the left side trials and

- foils on the right side trials, 8 assigned to be targets on the right side trials and foils on the left side trials.
- 3) **Pure AN**. *Used only in Experiment 2*. Each participant has new stimuli randomly chosen from the 2400 for every trial.
 - 4) **MIX4(VM,CM)**: *Used only in Experiment 1*: Each trial's study list has half CM and half VM stimuli. Both VM and CM have eight pictures randomly selected for each participant from the 2400. 4 CM pictures are assigned to be targets on the left and foils on the right, the other 4 CM pictures are assigned to be targets on the right and foils on the left. The eight VM pictures are randomly assigned each trial to side and target/foil.
 - 5) **MIX8(AN,CM)**: *Used only in Experiment 2*: Each trial's study list has half CM and half AN pictures, the CM pictures consisting of 16 stimuli randomly selected for each participant from the 2400. 8 CM pictures are assigned to be targets on the left and foils on the rights, the other 8 CM pictures are assigned to be targets on the right and foils on the left. The AN stimuli are new every trial, randomly selected each trial for each participant from the 2400.
 - 6) **MIX4(AN,CM)**: *Used only in Experiment 2*: Each trial's study list has half CM and half AN pictures, the CM pictures consisting of 8 stimuli randomly selected for each participant from the 2400. 4 CM pictures are assigned to be targets on the left and foils on the rights, the other 4 CM pictures are assigned to be targets on the right and foils on the left. The AN stimuli are new every trial, randomly selected each trial for each participant from the 2400.
 - 7) **AT**: *Used only in Experiment 2*: Pure VM and Pure CM conditions alternate every trial, each with a different set of 16 stimuli selected randomly for each participant from the 2400. The results for the VM and CM AT conditions did not match the corresponding pure VM and CM results, nor the MIXED results for those conditions. Rather it seemed that the participants used a mixture of strategies, a mixture that seemed to vary across participants. Because we are presenting a coherent account of a large set of results with a model assuming a common strategy and choice of criteria for all groups, conditions, and participants, we do not analyze and model the AT results.

Participants.

Participants were undergraduate students of Indiana University who received course credits as requirements in an introductory psychology course. They gave consent and were allowed to leave at any time without penalty. The study and participation were approved by the Indiana University IRB.

There were 156 participants in total in experiment 1. Two were deleted from analysis for obvious failures to obey instructions. The numbers in each condition were thus 28(CM), 28(VM), 32(MIX), 32(MIX8(VM,CM)), and 34 (AT). There were 204 participants in total in experiment 2. Two participants were deleted from analysis for obvious failures to obey instructions. The numbers in each condition were thus 34(CM), 35(VM), 33(AN),

32(MIX8(AN,CM)), 34(MIX4(AN,CM)), and 34 (AT). The experiment took about 45 minutes for each participant to complete.

[Conditions used in Nosofsky et al. (2021):

This study used the standard design with all study pictures and test pictures presented at fixation: The five groups were: **Pure AN**; **Pure VM**; **Pure CM**; **MIX1** (VM, CM and AN in equal numbers -- Cohort 1); **MIX2** (VM, CM and AN in equal numbers -- Cohort 2). 18 pictures were used for the pure VM conditions. The pure CM conditions used 9 targets and nine foils. In the mixed conditions the VM pictures were chosen from a set of 6 pictures, and the CM targets and foils were chosen from sets of three each. Details may be found in the published article.]

RESULTS

Figures 2a and 2b for error rates and 2c and 2d for correct response time medians show results for the eight groups (and different conditions) of the present studies and the five groups (and different conditions) of the published study, comprising 376 distinguishable conditions/data points for accuracy and another 376 for response time medians.

Fig 2a

RT-Lag: MIX conditions

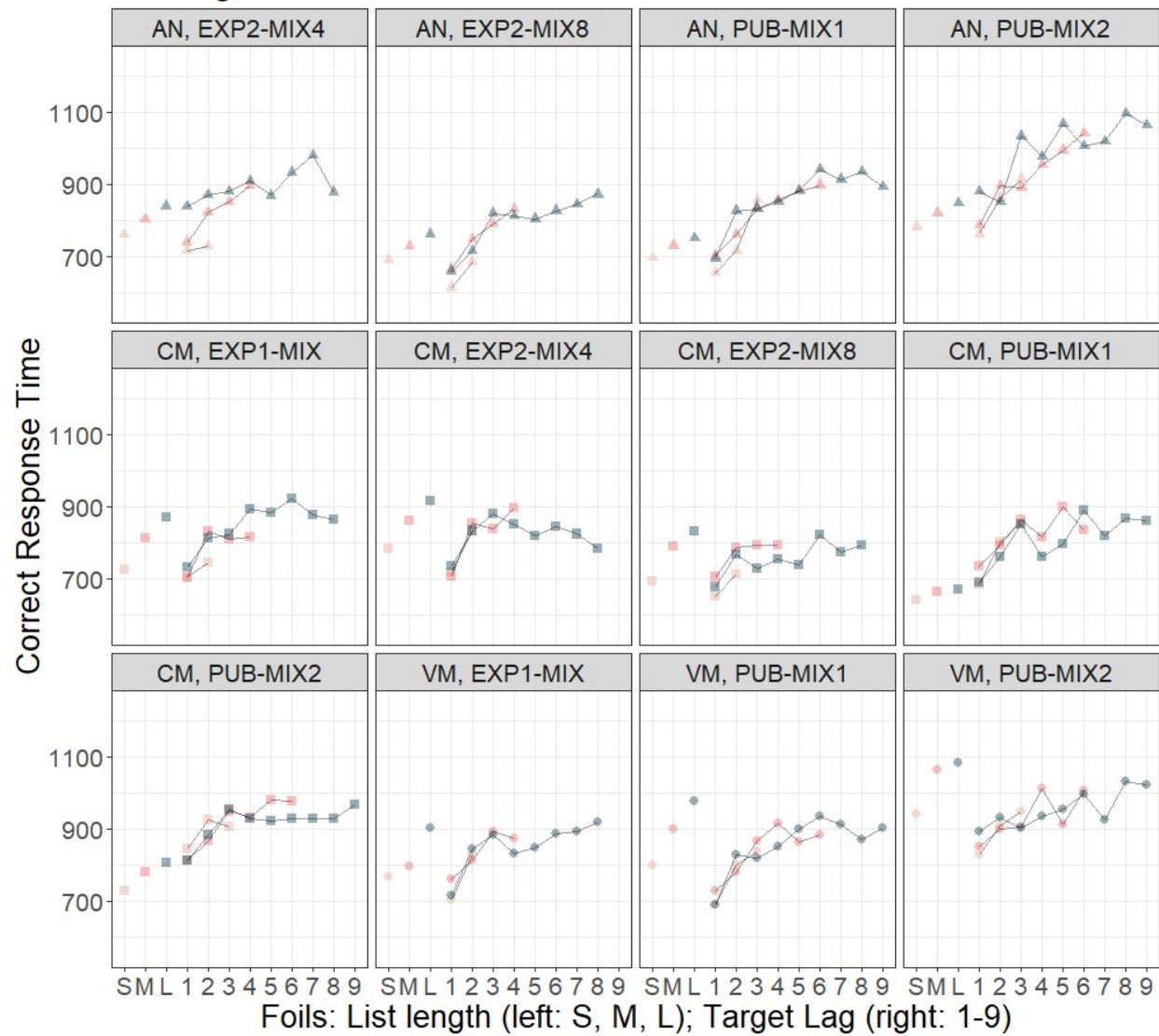


Fig 2b

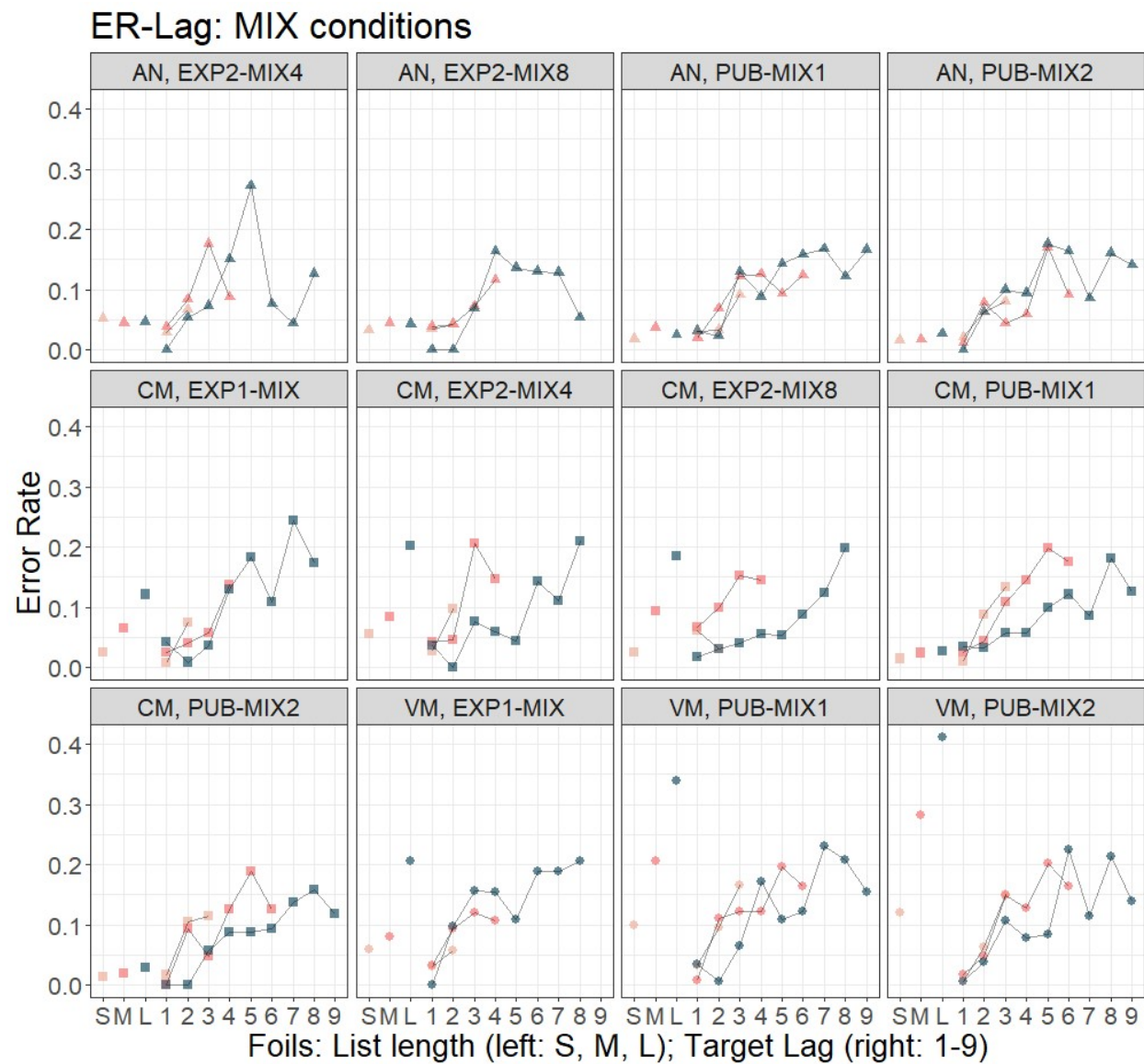


Fig 2c

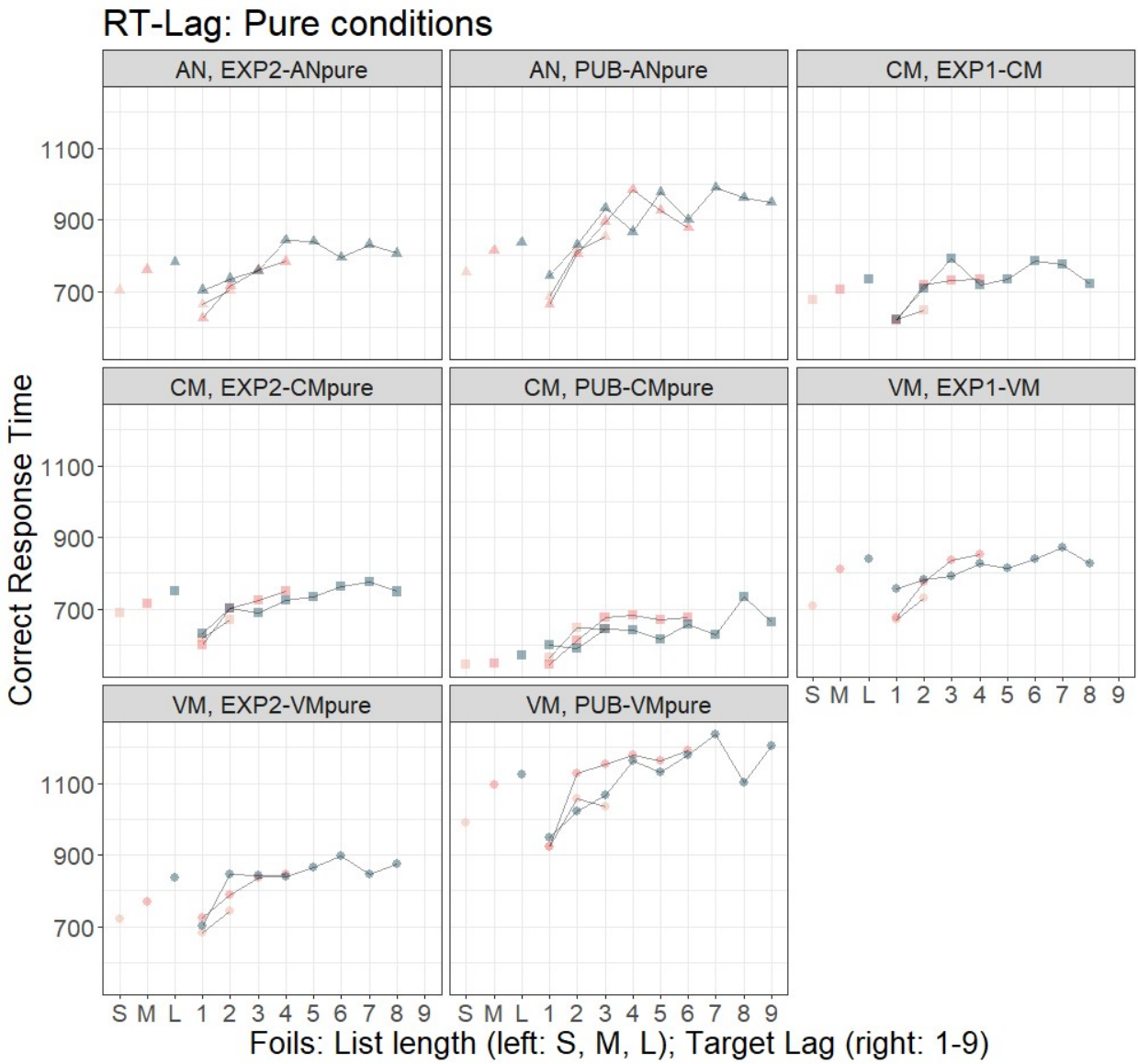


Fig 2d

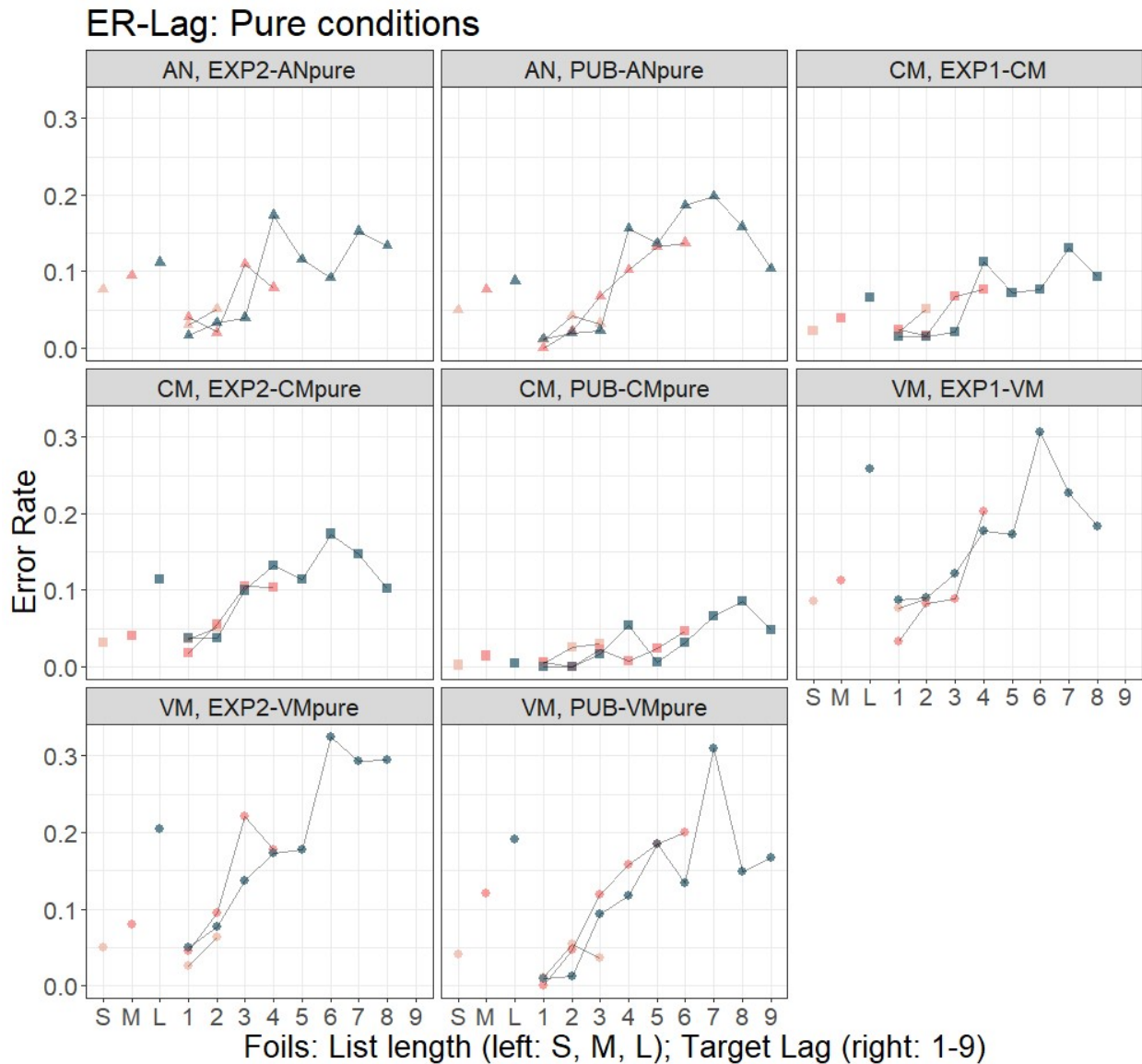


Figure 2 a,b,c,d: PUB refers to Nosofsky et al., 2021; EXP1 and Exp2 refers to present studies. EXP1-MIX refers to MIX4(CM, VM), EXP2-MIX4 refers to MIX4 (CM, AN), and EXP2-MIX8 refers to MIX8 (CM, AN). Each panel shows results for different list lengths. The lag results for targets for the three list lengths are superimposed on the right portion of each panel, the longest length being on the right end. There is a single foil result for each of the three list lengths; these are on the left of each panel, L indicating the longest list length, M the intermediate length and S the shortest length. Triangles are used for AN, squares for CM and circles for VM. Pink points are used for list lengths 2 or 3, red points for lengths 4 or 6 and blue points for lengths 8 or 9 (the target lags for different list lengths lie more or less atop each other). Figure 2a: Error rates as a function of lag and set size for the pure conditions of all studies; Figure 2b: Like Figure 2a, but for the mixed conditions. Figures 2c and 2d: Like Figures 2a and 2b but for median response times for correct responses.

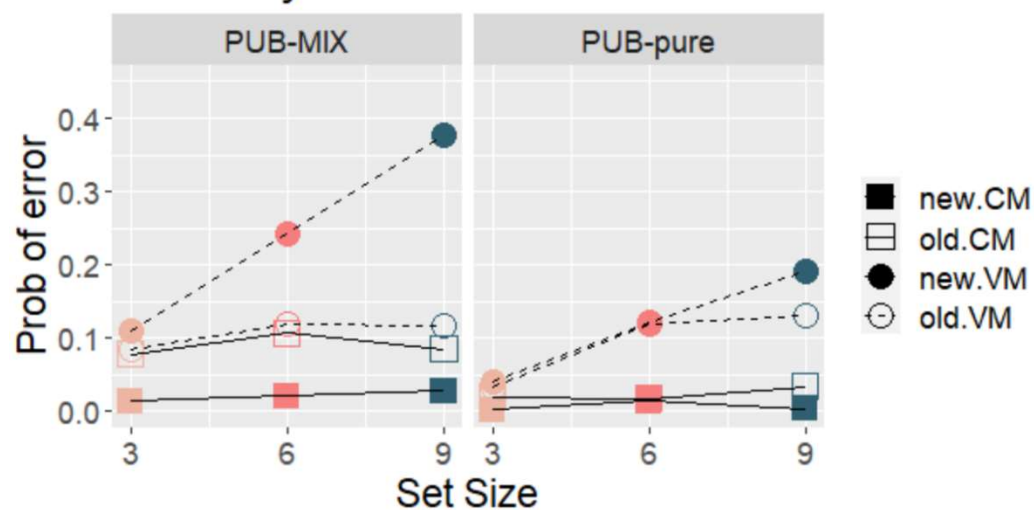
Figure 2a,b,c,d about here

In Figure 2 the many panels and their size make it hard to ascertain the exact values of the graphed points. An **Appendix** in the archived report gives the values for every graphed point in Figure 2, including the number of observations for each accuracy point (the same number applies for each RT point), and for response times the standard deviation of the medians across the participants (the average of the medians is shown in the various figures). That appendix also gives the model predictions for each point.

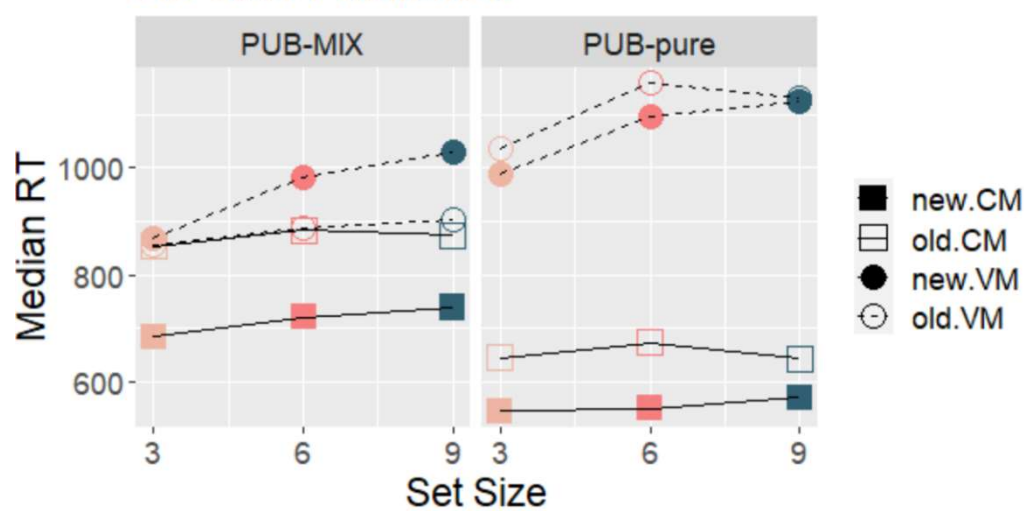
The many results shown in Figure 2 are difficult to comprehend piece by piece without interpretation by the modeling. Here we highlight a few major empirical results.

CM is Superior to VM: Figures 3a,b,c,d give target and foil results for CM and VM as a function of set size. The superiority of CM is rather obvious for the published study, and is present but less obvious for the present studies. These comparisons are of greatest interest for the mixed conditions where the same strategy is likely used for storage and retrieval for CM and VM pictures. The CM advantage in the present studies must be due to learning because the paradigm rules out other possibilities. Conservative sign tests show CM superiority in all conditions, particularly including the mixed conditions: Combining hits and correct rejections, the two-sided sign test between VM and CM for combined data of MIX1(CM,VM,AN) and MIX2(CM,VM,AN) for the published experiment is $p < .0001$ for both correct RT and accuracy. The two-sided sign test between VM and CM for the mixed conditions for the present experiment MIX4(VM, CM) is $p=0.004$ for accuracy and for $p=0.009$ for correct RT. Comparison of AN to CM and VM is best discussed after presentation of the model.

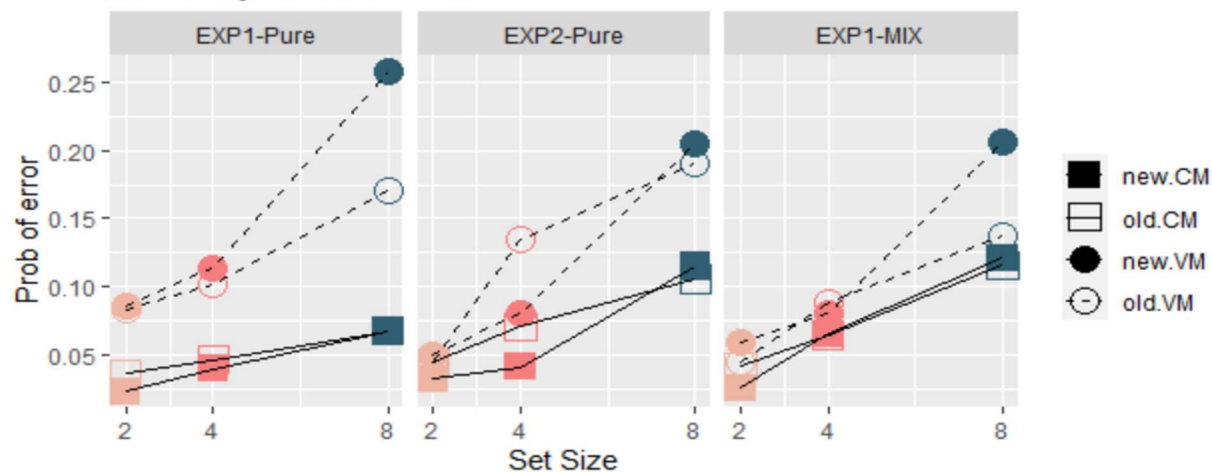
Accuracy obs: Published



RT obs: Published



Accuracy obs: Present



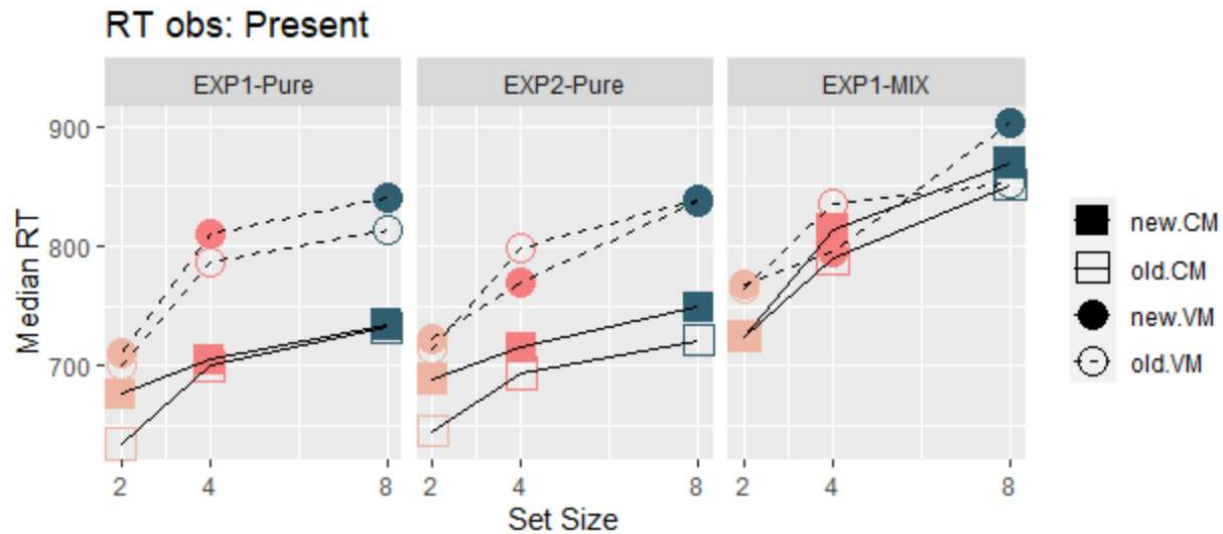


Figure 3: Observed data. Comparison of CM (solid lines) to VM (dashed lines) as a function of list length, for targets ('old'; open points) and foils ('new'; filled points). 3a: Error rates published study. 3b: Median correct response times published study. 3c: Error rates present studies. 3d: Median correct response times, present studies.

Figure 3a,b,c,d about here

Set-size and Target Lag Effects: For AN, CM and VM, as shown in Figures 2, 3, and 4, longer lists have lower performance both for accuracy (both hits and correct rejections) and response time for correct responses. All these results are very strong and reliable, obviating the need for statistics. The target set-size effect is due to longer lists having poor performance at long lags. At a given lag there is too much noise in the data to produce any significant difference in the target lag results as a function of list length (although the model presented later predicts a small decrease in performance for longer lists).

Mixed vs Pure: These results are complex, and are best discussed after presentation of the model.

MODELING

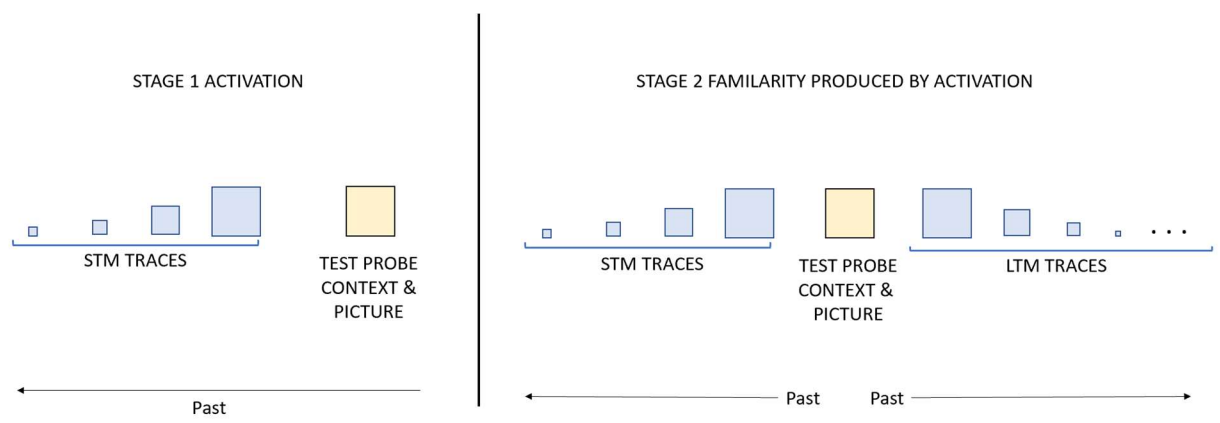
The primary goal of the present research was an understanding of the main processes of short- and long-term memory that operate in probe recognition. We aimed to achieve this goal by producing as simple, coherent, and justifiable a model as possible. We developed a new model because the model in Nosofsky et al. (2021) has certain unsatisfactory properties. We do not want to dwell on these but outline them briefly later in this report. The new model is termed SLR22 (standing for STM LTM Retrieval 2022). The model in the 2021 paper is similarly termed SLR21. SLR22 accounts for the patterns of findings for both the published study and the present studies, using a common model for both. Due to the large number of groups and conditions, and the large amount of data, the critical information from which we infer the processes at work lies not in statistically significant differences of some data points from others, but rather in the ability of a simple and coherent model to explain the data patterns.

The published article used a variant of the EBRW model originally used for categorization experiments by Nosofsky and Palmeri (1997). Because we are fitting both the published 2021 data and our new data with a single model, we again use the EBRW model framework. A detailed description of SLR21 can be found in the 2021 publication. A note on terminology: SLR22 incorporates many of the same processes as SLR21 but differs in ways that makes it confusing to use the same notation. Thus SLR22 uses conventional notation, with parameters notated by Greek letters presented in ***Bold Italic***.

SLR22 includes both a process of learning, and a process of decision based on familiarity.² "Learning" could be interpreted as storage of responses in traces in long-term memory (e.g. Gardiner & Java, 1990; Raaijmakers & Shiffrin, 1980, 1981) or as a matter of developing knowledge from events (Nelson & Shiffrin, 2013) and using that knowledge to govern the step direction of the random walk that is used in EBRW. The learning rate is expected to vary due to the varying difficulty and amount of training of the CM pictures in different conditions.

The "familiarity" process is based on a sum of activations of short-term traces. (e.g. Gillund & Shiffrin, 1984; Malmberg, 2008; Nosofsky, 2016; Lin & Oberauer, 2022), plus, a sum of activations of long-term traces (e.g.; Shiffrin & Steyvers, 1997, 1998; Dunn, 2008; Cox & Shiffrin, 2017). The amount of activation of each trace is determined by a match of the item and context in the probe to the item and context in the trace. Critically, the amount of trace activation is the same whether the trace encodes a target or foil response. This type of familiarity has been used on many occasions to model long-term recognition; among other findings it helps explain list length effects (e.g. Gillund & Shiffrin, 1984; Shiffrin & Steyvers, 1997, 1998) and the harm caused by earlier testing of other items (e.g. Criss & Shiffrin, 2004; Criss, Malmberg, & Shiffrin, 2011). That this type of familiarity is needed was shown convincingly by Nosofsky, Cao, Cox, and Shiffrin (2014). The production of familiarity occurs in two stages as illustrated in Figure 5, the idea being that the short-term traces are accessed faster than the long-term traces.

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² These two processes were assumed also in the modeling of Nosofsky et al., 2021, but implemented in a different way.

Figure 5: Familiarity is calculated in two stages, the first summing activation of short-term traces only, the second summing activations of both short- and long-term traces. The squares descending in size into the past represent the decreasing activation of older traces.

Figure 5 about here

Both learning and familiarity are modeled by evidence accumulation characterized as a random walk, with probabilistic steps (of evidence) accumulating until an OLD or NEW boundary is reached, at which time the response corresponding to that boundary is given. At each time epoch the probability, p_{TS} , of a step toward the **target** boundary given by Eq. 1a for a test of a target and 1b for a test of a foil. The probability of a step toward the **foil** boundary is $1.0 - p_{TS}$. In these eqs. S_{kT} and S_{kF} give the summed amount of activation of short-term traces (k is the list-length). The direction of the step is given by a function relating the activation values to a criterion value Φ .

$$p_{TS}(T) = \frac{S_{kT} + L_T + \Lambda_j}{S_{kT} + 2L_T + \Lambda_j + \Phi} \quad (1a)$$

$$p_{TS}(F) = \frac{S_{kF} + L_F}{S_{kF} + 2L_F + \Lambda_j + \Phi} \quad (1b)$$

Eqs. 1a,b are used to generate the predictions for both accuracy and response time for every one of the 376 conditions in both experiments. Because learning and familiarity are in the same equations, these forces jointly determine the step direction on each trial^{f2}. The variables in these equations depend on many parameters that will be specified below but it is important to note that the values of the parameters are held mostly constant across all the conditions. SLR22 provides an excellent *qualitative* account of the results from both studies. We were pleasantly surprised that the predictions were also reasonably close *quantitatively*. We believe the new model has justifiable assumptions, and justifiable parameter values.

f2: Footnote: It would be equally plausible to assume a mixture model in which either learning or familiarity occurs on each trial with some probability. It seems likely both approaches would produce similar predictions.

Definition: Traces that were stored due to study or test of the current test picture are termed ‘matching’ and traces that were stored due to study or test of a picture different from the current test picture are termed ‘mismatching’. This applies to traces stored on the current trial or any preceding trial.

Evidence Accumulation: Eqs. 1a,b incorporate two kinds of evidence that accumulate, one based on learning that occurs in CM, represented by the Λ term in the equations, and the other based on familiarity, represented by the S and L terms in the

equations, that operates in all conditions. Evidence accumulation models have probably been used over 1000 times since the 1960s (e.g, Laming, 1968; Link and Heath, 1975; Ratcliff, 1978; Vickers, 1979). Such models are most often used to explain data with both accuracy and response time measures. These have innumerable forms but have been highly successful and the present use needs no elaborate justification. In SLR22, evidence starts at zero. A unit step is taken either toward a target boundary or a foil boundary, and this continues until one of these boundaries is reached. Note: The closer is p_{TS} to 0.5, the more random and slow is the response.

The familiarity evidence is based on summed activation of short-term traces (S) and long-term traces (L), based on strength of trace and similarity to the test probe. When a target is tested one of the short-term traces *matches* the test item, producing strong activation, and the other traces *mismatch*, producing weaker activation. When a foil is tested all of the short-term traces *mismatch*. This is a primary basis for above chance recognition performance in all conditions.

Activation of long-term traces varies with conditions, but not current trial list length. Long-term traces that mismatch are activated very little due to the facts that the probe picture does not match the picture encoded in the trace, and the context used in the probe does not match the context stored in the trace. It is therefore reasonable to ignore mismatching long-term traces. Let L_+ be the total amount of activation of long-term *matching* traces that were stored due to prior trial *testing*. Let L_- be the total amount of activation of long-term *matching* traces that were stored due to prior trial *study*. The subscripts + and – are used because, as explained later, tests produce stronger traces than study. The sum, $L_+ + L_-$, is denoted L_T for tests of targets and L_F for tests of foils. L_T equals L_F for all groups and conditions except for the CM conditions in Nosofsky et al. (2021). In those conditions $L_T > L_F$ because L_- (traces stored due to study) is larger for tests of targets than foils.

The criterion Φ in Eqs. 1a,b was kept constant across groups and conditions.

Special cases of equations 1a and 1b apply to AN and VM:

For tests of AN targets or foils, there are no long-term matching traces and no learning:

$$p_{TS}(T) = \frac{S_{kT}}{S_{kT} + \Phi} \quad (2a)$$

$$p_{TS}(F) = \frac{S_{kF}}{S_{kF} + \Phi} \quad (2b)$$

As S_{kT} gets larger, and S_{kF} gets smaller, there is a higher probability of getting to a boundary correctly, and doing so faster, so the difference $S_{kT} - S_{kF}$ can be a kind of measure of accuracy.

For tests of VM targets or foils there is no learning. Because $L_T = L_F$ both are denoted L in Eqs. 3a,b. They are equal because each picture is randomly made a target or foil as trials continue.

$$p_{TS}[T] = \frac{S_{kT} + L}{S_{kT} + 2L + \Phi} \quad (3a)$$

$$p_{TS}[F] = \frac{S_{kF} + L}{S_{kF} + 2L + \Phi} \quad (3b)$$

Equations 1 through 3 have the properties need to fit the data. Eqs. 3a,b for VM predict worse performance than Eqs. 2a,b for AN, because $L/2L$ moves p_{TS} closer to 0.5. Eqs. 1a,b predict CM to be better than AN and CM due to learning. In Eq. 1a, A_j is in both the numerator and denominator, so increases p_{TS} for target tests. In Eq. 1b, A_j is only in the denominator, so decreases p_{TS} for foil tests.

- 2. Strength Decay in Short-Term Memory:** Traces in short-term memory from the study list vary in strength, m , with the lag h :

$$m_h = h^{-\beta} + \alpha \quad (4)$$

This assumption predicts the strong recency effects seen in the target lag functions in many studies using rapid presentations (i.e. Monsell, 1978; Nosofsky et al., 2021). Studies by Sternberg (e.g. 1966) obtained different results, but used slower rates of presentation and test, allowing rehearsal and other strategies to distort the recency effects.

- 3. Stages of Processing:** The idea that perception operates in a series of stages, with simpler features perceived first, goes back over 100 years. A recent implementation of this idea used for memory retrieval in recognition (and recall) is found in Cox and Shiffrin (2017; see also Holmes, Trueblood, & Heathcote, 2016; Harding et al. 2021). We approximate stages of feature processing with just two stages, the first lasting for just one step of the random walk process. The first stage is assumed to activate only the then available traces in short-term memory and to use low-level physical features so AN, VM, and CM items are activated similarly. In stage 2, both short- and long-term traces are activated. Stage 2 continues until a response boundary is reached.
- 4. Similarity Differences:** Differences in activation of different types of short-term traces by different test probes are represented by differences in the similarity, Ω , between test probe and trace. The similarity values are listed as part of Table 2. Two stage 1 and seven stage 2 similarity parameters were estimated. These nine values were the same for all conditions, depending only on the picture types stored and used as a test probe.
- 5. Short-term Trace Activation:** The activation of the short-term trace t_h at lag h by test Item e_i is termed a_{ih} and is a product of its storage strength m_h and similarity Ω to the

test item:

$$a_{ih} = m_h \Omega \quad \Omega \leq 1 \quad (5)$$

S_k is the summed activation of all short-term traces for list length k :

$$S_k = \sum_h a_{ih} \quad (6)$$

6. Long-term Trace Strength

Because stimuli on each trial are presented rapidly, and short-term trace strength decreases rapidly with lag, we assume that long-term trace strength is mostly determined by processes occurring at test and during the period after test and prior to the next trial. If so, tests would have a stronger influence on long-term storage than study. Note that only one picture on each trial the one tested foil or target, gets this benefit from testing. Our assumption that testing produces stronger long-term traces than study is generally consistent with research in showing that testing provides more benefit for long-term retrieval than study (e.g. Roediger & Karpicke, 2006).

7. Context Change and Long-term Trace Activation:

The summed activation of long-term matching traces is assumed to vary in accord with the strength of the matches between a test picture and different long-term traces. This strength depends on the degree to which the context in the test probe and the trace match, and hence falls off rapidly for traces from older trials. As a simple approximation we calculate the *probability of a match on the prior trial* and let total activation be proportional to that probability. This probability is calculated separately for a trace stored due to either test or study, as mentioned in the previous paragraph. These probabilities are not estimates. They vary with group and condition as determined by the design of the studies, with values shown in **Table 1**.

	PT(E)	PT(U)	PF(E)	PF(U)
<u>SIDE VM PURE:</u>	1/16	7/24	1/16	7/24
<u>SIDE CM PURE:</u>	1/16	7/24	1/16	7/24
<u>SIDE MIX8-CM/AN CM:</u>	1/32	7/48	1/32	7/48
<u>SIDE MIX4-CM/AN CM:</u>	1/16	7/24	1/16	7/24
<u>SIDE MIX4-CM/VM CM:</u>	1/16	7/24	1/16	7/24
<u>SIDE MIX4-CM/VM VM:</u>	1/16	7/24	1/16	7/24
<u>PUB VM PURE:</u>	1/18	1/3	1/18	1/3
<u>PUB CM PURE:</u>	1/18	2/3	1/18	0
<u>PUB MIX VM:</u>	1/18	1/3	1/18	1/3
<u>PUB MIX CM:</u>	1/18	2/3	1/18	0

Table 1: Probabilities that the test item has a match on the prior trial.

Table 1 about here

The way probabilities in Table 1 are converted to activation uses this notation:

$p_+(T)$ and $p_-(T)$ are the probabilities of a prior trial matching trace stored due to testing and study, respectively, when a target is tested on the current trial.

$p_+(F)$ and $p_-(F)$ are the probabilities of a prior trial matching trace stored due to testing and study, respectively when a foil is tested on the current trial.

Let ϵ be the weight given to study ($0 < \epsilon < 1.0$, to represent lesser strength for study than testing). Then:

$$L_T = \pi[p_+(T) + \epsilon p_-(T)] \quad (7a)$$

$$L_F = \pi[p_+(F) + \epsilon p_-(F)] \quad (7b)$$

The value π controls the amount of long-term trace activation and therefore represents the weight given to long-term trace activation in comparison to short-term trace activation. $\pi [p_+(T)]$ and $\pi[\epsilon p_-(T)]$ represent L_+ and L_- for target tests. $\pi [p_+(F)]$ and $\pi [\epsilon p_-(F)]$ represent L_+ and L_- for foil tests. The values of π and ϵ were set equal for all groups and conditions.

8. Response times: The boundaries for the random walk are θ_T and θ_F for a target and foil responses respectively. Although these could well vary across conditions, their values were fixed.

Eq. 8 gives the expressions for the response time for the random walk process. The predicted time includes a ‘base time’, or ‘residual time’ τ that represents time for processes not explicitly modeled such as the motor responses required to press the response key. These were allowed to vary for the 13 participant groups.

$$RT_{RW} = (N_s)(\chi) + \tau \quad (8)$$

N_s is the number of steps needed to reach the boundary, and the subscript RW indicates this is the predicted response time for the random walk. The mean value of N_s for various conditions was determined by an approximation derived by Bussemeyer (1982).

9. Glitches: It is assumed that performance is a mixture of ‘regular’ trials and ‘guessing trials’ termed ‘glitches’ (e.g. Ratcliff & Kang, 2021; Clark & Merfield, 2021). The probability of a glitch is ξ with response time τ_ξ , with values fixed across groups and conditions. Thus:

$$P(C) = (\xi)(0.5) + (1 - \xi)[p(c)] \quad (9)$$

where $P(C)$ is the total predicted accuracy and $p(c)$ is the accuracy predicted by the random walk. Mean response time is:

$$RT = (\xi)(\tau_\xi) + (1 - \xi)RT_{RW} \quad (10)$$

Fitting the Data:

The parameter values shown in **Table 2** minimized a discrepancy between predictions and observations summed across all distinct data points in both studies (those in Figures 2a,b,c,d). The discrepancy in accuracy was a squared difference in probabilities, and the discrepancy in response time was a squared difference in seconds between predicted mean time and observed mean of medians across participants (mean of medians was used to eliminate distortions caused by occasional extremely long response times). However, to produce qualitatively good results the errors in accuracy were weighted two times more than errors in response time. The discrepancy for foils was weighted four times more than the discrepancy at a given lag (because there is much more data in most foil data points than most target lag data points). It is worth emphasizing that the only parameter values allowed to vary across groups and conditions were learning rates in the eight CM conditions and base times in the 13 participant groups.

Parameter	Explanation	Specific condition	Estimated value
Φ	Criterion		0.352
χ	Response time taken per step to make a decision		17.312
Ω	Short-term Similarity between pictures of specific kinds	VM and CM	0.155
		VM and VM	0.121
		CM and CM	0.208
		AN to AN	0.104
		VM and AN	0.142
		VM and AN	0.060
α	The basic value of decay		0.121
β	The rate of memory decay		2.199
ξ	Probability of making glitches		0.060
T_{ξ}	Time taken for making glitches		523.863
π	The amount of long-term trace activation		3.008
ϵ	The fraction of the strength of long-term traces due to testing		0.001
OLD	The old boundary of the random walk		8.00

NEW	The new boundary of the random walk		-2.303
Λ	Long-term learning of CM	MIX-E1	0.212
		CM-E1	0.409
		CM-E2	0.313
		MIX4	0.058
		MIX8	0.039
		MIX-PUB	0.363
		MIX-PUB	0.387
		CM-PUB	0.951
τ	The average base time to make a decision	MIX-E1	282
		CM-E1	249
		CM-E2	242
		VM-E1	256
		VM-E2	260
		MIX4	299
		MIX8	261
		AN-E2	265
		MIX-PUB	277
		MIX-PUB	337
		CM-PUB	217
		AN-PUB	304
		VM-PUB	409

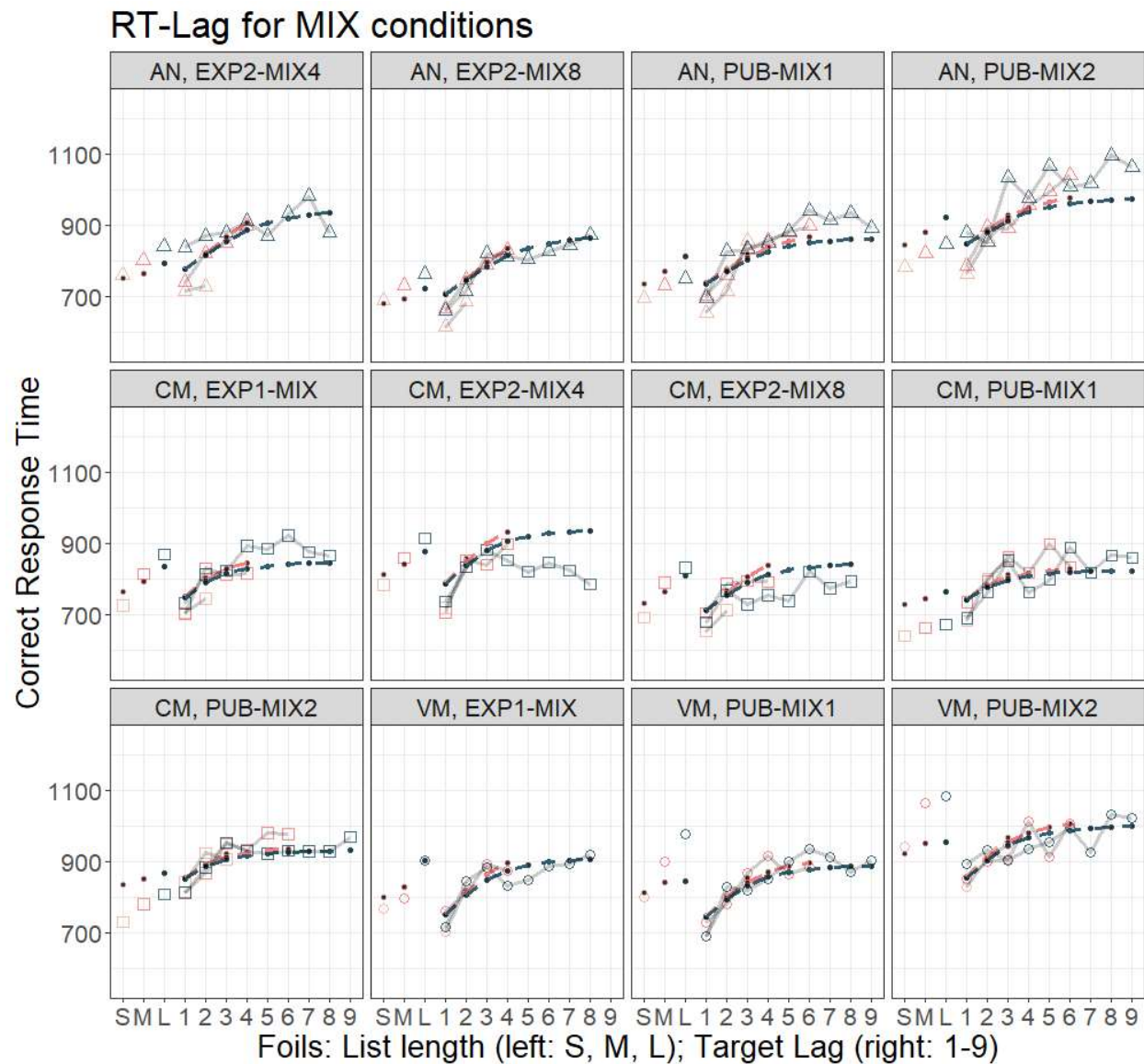
Table 2: Estimated parameter values³.

Table 2 about here

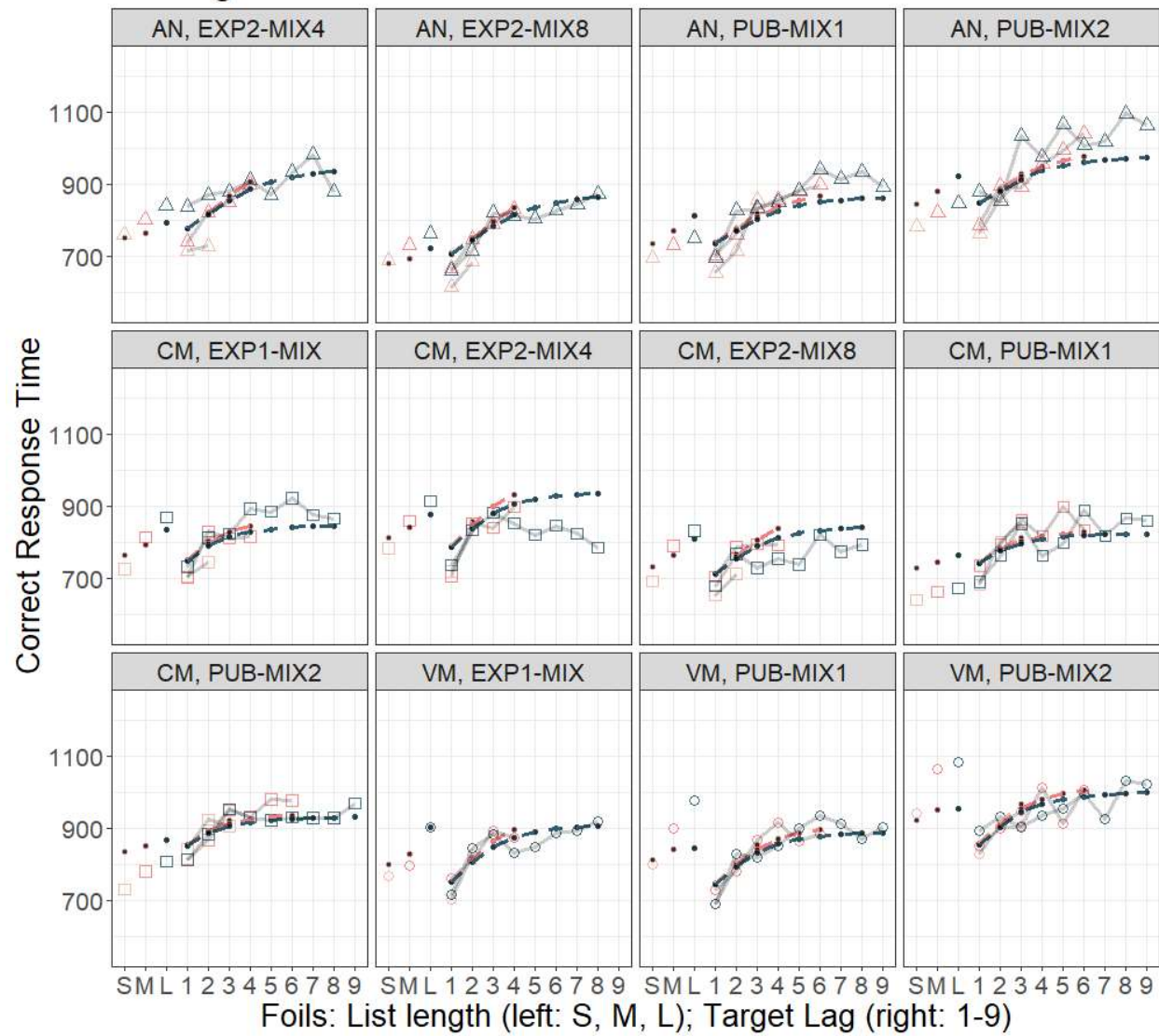
Predictions

³ These are close to the best fitting values, but may not be exact due to the limitations of the optimization program that was used.

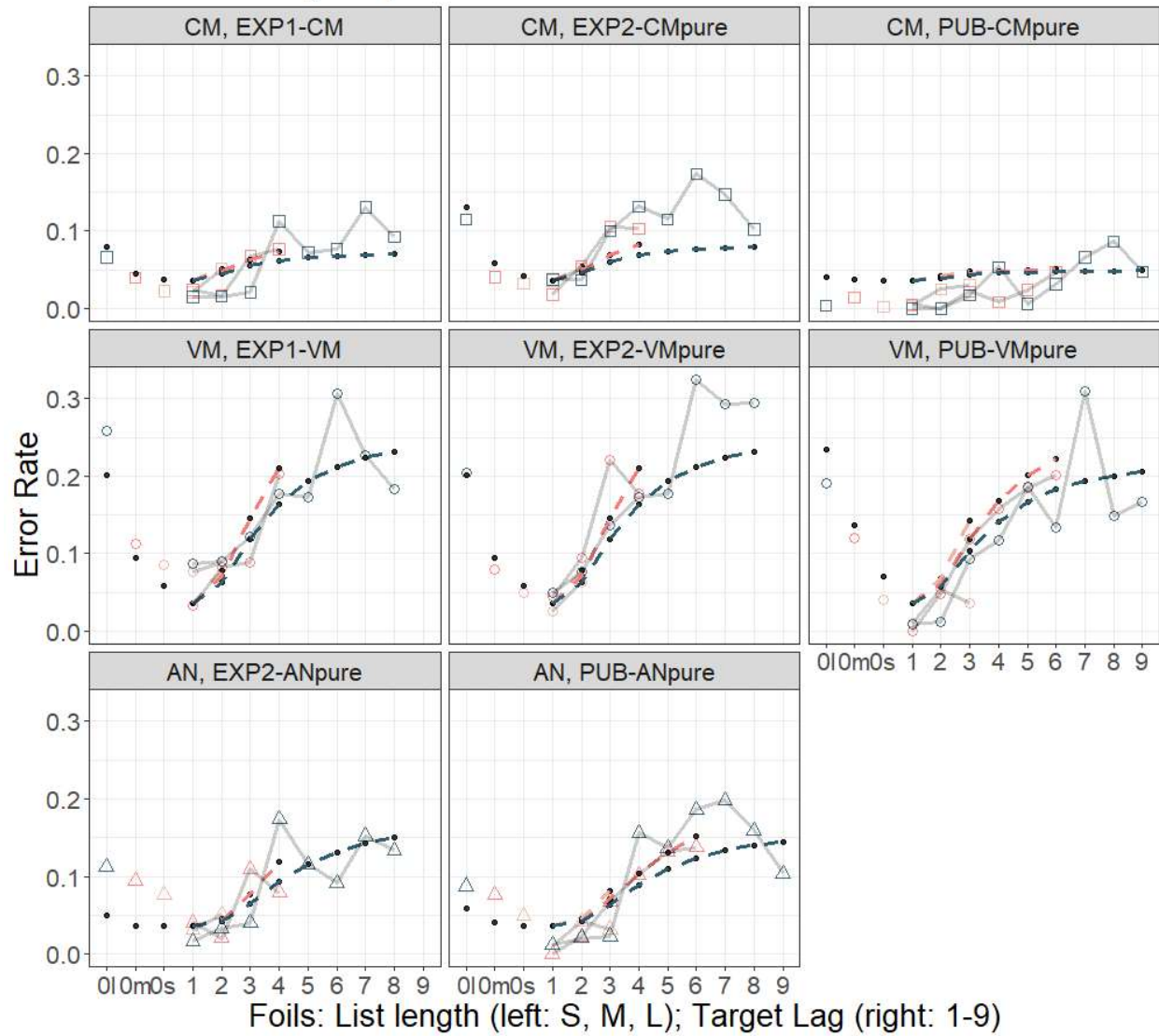
Figure 4a,b,c,d gives the predictions of the fitted model superimposed on the data of Figure 2a,b,c,d. This figure shows that the model qualitatively predicts the patterns of data quite well. (An **Appendix** in the archived report gives the predictions in numeric form). It is perhaps surprising that most of the predictions are also pretty close quantitatively, given that most parameter values were kept constant across the many groups and conditions.



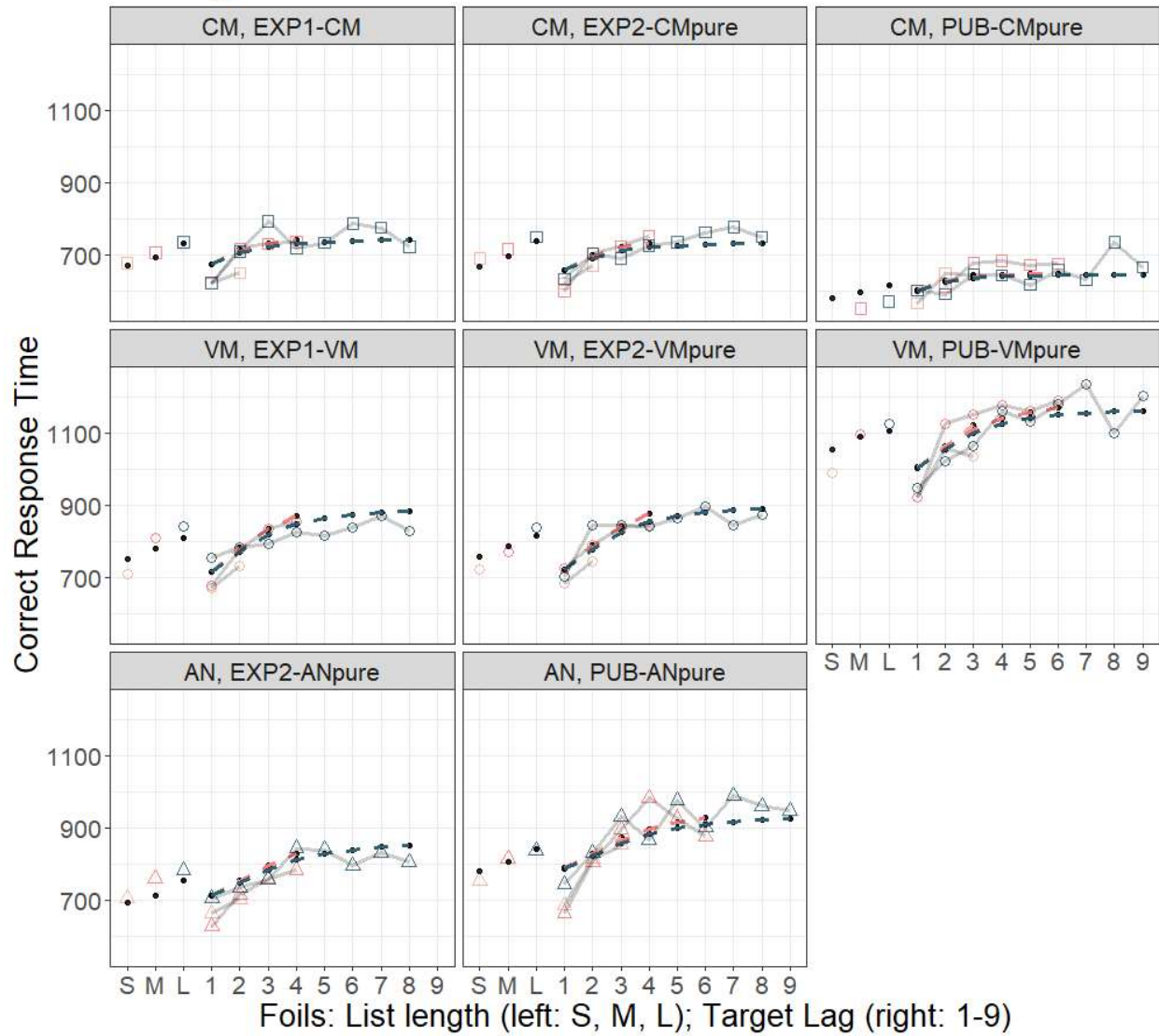
RT-Lag for MIX conditions



ErrorRate-Lag for pure conditions



RT-Lag for Pure conditions



ErrorRate-Lag for MIX conditions

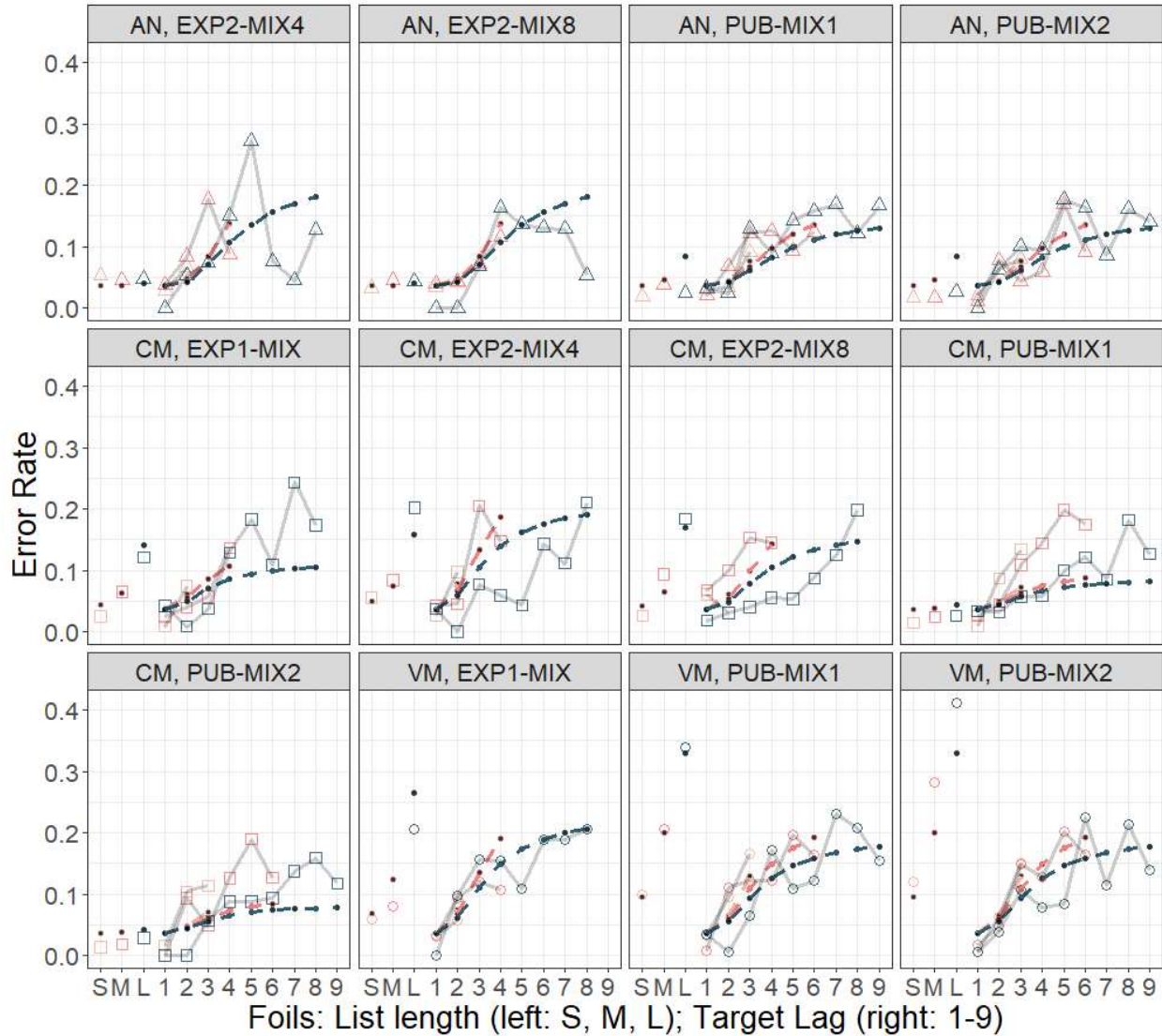


Figure 4a,b,c,d: Predictions shown as solid black points superimposed on the observed data. PUB refers to Nosofsky et al., 2021; EXP1 and Exp2 refers to present studies. EXP1-MIX refers to MIX4(CM, VM), EXP2-MIX4 refers to MIX4 (CM, AN), EXP2-MIX8 refers to MIX8 (CM, AN). Each panel shows results for different list lengths. The lag results for targets for the three list lengths are superimposed on the right portion of each panel, the longest length being on the right end. There is a single foil result for each of the three list lengths; these are on the left of each panel, L indicating the longest list length, M the intermediate length and S the shortest length. Triangles are used for AN, squares for CM and circles for VM. Pink points are used for list lengths 2 or 3, red points for lengths 4 or 6 and blue points for lengths 8 or 9 (the observed target lags for different list lengths lie more or less atop each other, although longer lists are predicted to have slightly inferior performance). Figure 4a: Error rates as a function of lag and set size for the pure conditions of all studies; Figure 4b: Like Figure 2a, but for the mixed conditions. Figures 4c and 4d: Like Figures 4a and 4b but for median response times for correct responses.

Figure 4a,b,c,d about here

There are a few quantitative mispredictions. One would be problem for any model: The target lag response time results for Exp. 2 MIX4 do not slow as lag increases. This is the one exception to a general rule of lower performance at longer lags. We have no explanation for this idiosyncratic result. Another quantitative misprediction occurs in two cases, both involving foil data and predictions for VM in mixed conditions: In the present experiment 1 foil errors were predicted to be too low; in one cohort but not the other of the published study, foil errors were predicted to be too high. Smaller mispredictions for these conditions can also be seen for response times. These cohorts were supposed to be replications, so of course a model assuming the same parameter values for both must fail.

The remaining quantitative mispredictions involve the foil predictions: Many are accurate but some predictions are too high and some too low. The parameter estimation programs weighted foil data four times that of target data; an even higher weight for foils would have produced better predictions for foils at the cost of worse predictions for targets. That issue aside, we could have allowed almost any of the parameters to vary by condition, and obtained a more accurate quantitative fit, but desired to demonstrate that a model with minimal flexibility would produce a good qualitative account. Indeed the pattern of the foil predictions were always in the correct direction. In general we judge the model to have produced an excellent qualitative account for the results and were surprised that the quantitative account was as good as it was.

Model Flexibility, and Parameter Variations

Although a plausible case could be made that each of Θ_T , Θ_F , Φ , ξ , Ω , π , λ , and τ should vary with participants, groups, conditions, and list length, we attempted to keep such variations to a minimum. We could not find a model that kept all parameter values constant, but were pleased to produce a model providing a good account when allowing only the learning rates λ in the 8 CM conditions and base times τ for the 13 groups to vary. There were excellent reasons why the learning rate should have varied across conditions, and as detailed shortly they varied in the way expected.

In evidence accumulation models generally, base times have been used as an adjustment for processes the theorist does not want to model explicitly (such as propensity to push the response keys more or less quickly), so we followed standard practice in allowing these to vary. Was it reasonable or unreasonable to do so for the 13 participant groups? Had the base times instead been fixed at a single value, but all other parameter values left unchanged, the accuracy predictions (which were quite good) would not have changed. In addition the predicted patterns of response times would have been qualitatively accurate, save for a shift upward or downward of the predictions in each of the response time panels.

For a data set this large with this many groups and conditions, with both measures of accuracy and response times, we regard the complexity of SLR22 as relatively low. Of course such a data set did require a fair number of processes and parameters to represent those processes. The model used 40 freely fit parameters. 13 were base times varying by group, and 8

were the learning rates varying across the different CM conditions. The remaining 19 parameters had values fixed for all groups and conditions.

Estimated Parameter Values

To make the case that the present model provides a reasonable and justifiable account of the core processes at work in these studies, it is important that those parameters whose values are interpretable theoretically have reasonable values, particularly in relation to each other.

Short-term Decay: The value of β , the short-term decay rate, was about 2.2 and produces the predicted lag curves that conform closely to those observed. The estimated value of α , the asymptotic value of short-term decay, was about .12. This value is hard to interpret, presumably representing some combination of a flattening of short-term loss at long lags, and a contribution from long-term memory.

Learning in CM Conditions: In Nosofsky et al. (2021) both the assignment of pictures to target or foil and to the response key were consistently mapped. The present study used a more complex design, and only the response key was consistently mapped. Thus the published study learning rate should have been faster and was: $[\text{.95} > \{.41, .31\}]; [\{.39, .36\} > \{.21, .06, .04\}]$. In all studies learning should have been slowed in the mixed conditions relative to the pure conditions because only one third or one half of the pictures on each trial were consistently mapped. The pure conditions in fact had faster learning: $[(.95) > (.39, .36)], [\{.41, .31\} > \{.21, .06, .04\}]$. We argued earlier that learning might have been faster in Mix4 than Mix8; the estimates were in this direction but the effect was tiny: $[\{.21, .06\} > .04]$.

A note about learning: The model represents learning as an increase in the probability of taking a step toward the correct response boundary. This representation was used because it had been used in the EBRW modeling in the published article and earlier ones. We investigated an alternative in which learning is represented by a probability of recalling a correct response, thereby bypassing the usual familiarity process. This mixture model produced predictions similar to those we have shown in this article.

Similarity Differences between AN, VM, CM Pictures: These were expected to differ and did, but we had no strong a priori predictions of the way these should differ, other than the prediction that match similarity should be higher than mismatch, which was the case: Stage 1: $[2.4 > 1.4]$; Stage 2: $[\{3.5, 1.0\} > \{.21, .15, .14, .12, .10, .06\}]$.

Glitches: The probability of glitch occurrence should be low and was: (.05).

Long-term Trace Strength: Prior research suggested that tests should produce stronger traces than study, and the value of ϵ , .001, showed this was the case for the present studies. This value was lower than expected. However, when we tried fitting the model using some higher values for ϵ , the qualitative fit of the model was not substantially worse. There seem to be correlations between model parameters such that there are trade-offs, including a trade-off of increases in the strength of traces due to study with decreases in the strength of traces due to tests.

Base Times: Although we had no strong a priori predictions, the base time estimates seem quite reasonable. The five groups with mixed conditions had slower base times than the eight groups with pure conditions, with one exception for AN in the published study. For the pure conditions the three VM base times were all slower than the three CM base times. Thus the base times seem to be mostly ordered by task difficulty for the 13 groups.

The remaining parameter values are hard to interpret, mostly because they emerge from interacting processes in EBRW. It should be noted that the parameter space has a number of local minima with similar goodness of fit, with parameter magnitudes varying, but almost always maintaining the ordinal properties listed above.

Model Comparison

There are uncountable numbers of model types and variants that could conceivably be employed in attempts to fit the present data set. We suspect that many types of evidence accumulation models could do a decent job if employing the processes of our model and suitably adjusted. We note only that the process of producing the present model was long and arduous, probably due to the many groups and conditions, and the complexity of the results.

An extant model deserving discussion is that of Nosofsky et al. (2021): SLR21 was used to predict the data from five groups (a subset of the 13 groups modeled in this article). SLR21 had the merit of providing a good quantitative account of a complex data set, used many of the same processes of SLR22, and led to many of the same conclusions. We were led to seek a better model because the conclusions reached using SLR21 were undercut by estimated values of key parameters that were not reasonable:

- 1) In the Mixed condition there was a three times larger learning rate for NEW items than the combined learning and familiarity strength for OLD items, even though old CM items were seen more often and should have been more familiar and should have been learned better.
- 2) The pure CM condition required a huge learning strength for NEW test items and a very small combined strength for OLD test items, again the opposite of expectations.
- 3) In the mixed conditions there were equal contributions of long-term traces in VM and AN (equal for both targets and foils), but AN has no matching long-term traces while VM has many.

SLR22 resolved these problems. All the pure and mixed conditions from both the five groups of Nosofsky et al. (2021) and the eight groups of the current studies were fit with the same model with most parameter values held fixed across all groups and conditions. The processes in the model are based on those that have received good support in prior research. The estimated values of the parameters were reasonable and met expectations. We note that the conclusions reached here match many of those reached in the published article; thus we do not argue that they were wrong, but rather that they are now much better justified.

Conclusion

The present research and modeling make it clear that long-term traces are activated and strongly affect short-term recognition: In VM conditions, prior trials are equally often targets and

foils, and thus the traces of their pictures cannot help short-term recognition when activated. The poor performance in VM relative to AN makes it clear that long-term activation is actually harmful. We suggest the harm is due to noise caused by variability in long-term activation, something represented in the modeling by the 2 in the denominator of Eqs. 1a,b. One might believe the participant would use context to focus retrieval so strongly on the traces of the recent list that long term traces would not be activated, eliminating the harm. However, the results show that this is not possible.

The situation is quite different in CM conditions: When a target is tested all long-term traces that match the test picture encode target responses, and when a foil is tested all long-term traces that match the test picture encode foil responses. One might think that retrieval of those traces would therefore help performance. However, that help would require a form of recall, by which the contents of the traces were retrieved and used. Our model of familiarity excludes such recall: Activation depends on the match of context and picture in the probe to context and picture in the trace, but not upon the response encoded in the trace. Thus our model has a separate process of learning that is a form of recall. We think it best to interpret such learning as the development of long-term knowledge (e.g. Nelson & Shiffrin, 2013): In essentially every real-life learning situation it is consistent training that produces knowledge, and consistent training of the contingency between stimulation and action that leads to improved performance.

The particular way we combined the two processes of learning and familiarity in SLR22 may or may not be best, but the way the model produced an excellent account of a very large and complex data set with many conditions gives some support to the modeling. The data were from 13 groups of participants, and 376 conditions that included measurements of both accuracy and mean response times. The conditions included three ways that the pictures varied across trials: 1) new pictures studied and tested every trial (AN); 2) every trial used a selection from a small set of pictures, and a given picture was on half the trials a target and on half the trials a foil (VM); 3) every trial used a selection from a small set of pictures but half the pictures were targets on every trial and the other half foils on every trial (CM; this condition allowed learning of the responses to each picture to take place). Conditions included ‘Pure’ conditions in which a given picture type occurred on all trials (e.g. every trial was either AN, VM, or CM), and ‘Mixed’ conditions in which each trial contained equal numbers of AN, VM, CM pictures (Nosofsky et al. 2021) or equal number of just two of these picture types (the present studies). Conditions included variations of study list length, 3, 6, or 9 in Nosofsky et al. (2021) and 2, 4, or 8 in the present studies. When targets were tested the position of the target in the study list (‘lag’) was varied systematically. Finally, Nosofsky et al. (2021) used standard probe recognition with all trials centered at fixation; in contrast the present studies used a new paradigm with trials alternating presentations on the left and right side of fixation in such a way that targets and foils had equal numbers of exposures and equal familiarity.

The heart of the modeling is surely the two processes of learning and familiarity. The model of course incorporated a large number of other processes. We believe each was justified not just because the model produced a good qualitative account of the patterns of data from many conditions, but also on the basis of reason and past research:

1. Study of an item causes a memory trace to be placed in short-term memory and another in long-term memory, both containing item and context information.

2. Test of an item causes a strong memory trace to be placed in long-term memory.
3. Testing an item requires a probe of memory with item and context information.
4. Probing memory with item and context information causes activation of traces in short- and long-term memory.
5. Context changes as trials continue.
6. Short-term memory has limited capacity and thus traces in short-term memory have limited duration. In the absence of rehearsal of earlier items, their short-term traces are reduced in strength, and/or leave short-term memory, as other items are subsequently studied.
7. Activation of either a short- or long-term trace is determined jointly by the strength of the trace and the similarity of the test probe to the trace.
8. Test stimuli are perceived in stages, starting with low level features, the early stages mainly activating short-term traces.
9. Recognition decisions are generally based on ‘familiarity’, a process based on amount of trace activation but not recall of the contents of the traces.
10. The accuracy and timing of responses are well accounted for by dynamic processes of evidence accumulation.
11. When stimuli are consistently mapped to response across trials, response learning occurs, represented by recall of information stored in long-term memory.

Using these assumptions we were able to account quite well for the complex and rich pattern of results with what we regard as a quite simple and coherent model, given that the parameter values representing the many processes at work were kept mostly the same for all groups and conditions in both the published study and the present studies.

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Appendix

Experiment	Probe	SS	Lag	Old	cRT	Pred cRT	cRT SE	Error Rate	Pred Error rate	Error Rate SE
EXP1-CM	CM	2	0	new	676.23	668.71	32.20	0.0229	0.0381	0.0078
EXP1-CM	CM	2	1	old	620.75	674.58	28.96	0.0211	0.0365	0.0077
EXP1-CM	CM	2	2	old	648.84	713.96	27.93	0.0509	0.0506	0.0174
EXP1-CM	CM	4	0	new	705.30	691.62	26.52	0.0395	0.0449	0.0129
EXP1-CM	CM	4	1	old	619.61	673.82	34.30	0.0239	0.0365	0.0109
EXP1-CM	CM	4	2	old	717.20	709.98	34.17	0.0164	0.0476	0.0096
EXP1-CM	CM	4	3	old	729.98	731.31	32.40	0.0676	0.0631	0.0268
EXP1-CM	CM	4	4	old	733.77	741.89	27.89	0.0764	0.0731	0.0314
EXP1-CM	CM	8	0	new	734.36	731.74	27.95	0.0664	0.0797	0.0208
EXP1-CM	CM	8	1	old	621.14	672.86	27.73	0.0153	0.0364	0.0153
EXP1-CM	CM	8	2	old	709.38	705.15	39.27	0.0155	0.0447	0.0123
EXP1-CM	CM	8	3	old	792.50	722.91	43.63	0.0208	0.0555	0.0118
EXP1-CM	CM	8	4	old	718.52	731.42	33.25	0.1128	0.0622	0.0514
EXP1-CM	CM	8	5	old	732.66	735.80	36.43	0.0718	0.0661	0.0427
EXP1-CM	CM	8	6	old	786.40	738.25	38.76	0.0767	0.0683	0.0340
EXP1-CM	CM	8	7	old	774.44	739.74	43.31	0.1307	0.0698	0.0480
EXP1-CM	CM	8	8	old	721.90	740.70	36.34	0.0927	0.0707	0.0329
EXP1-MIX	CM	2	0	new	724.73	764.10	30.46	0.0254	0.0438	0.0111
EXP1-MIX	CM	2	1	old	704.27	750.96	27.90	0.0083	0.0367	0.0060
EXP1-MIX	CM	2	2	old	744.87	801.67	27.50	0.0747	0.0603	0.0185
EXP1-MIX	VM	2	0	new	767.57	798.98	33.84	0.0588	0.0693	0.0156
EXP1-MIX	VM	2	1	old	702.11	752.40	31.83	0.0301	0.0369	0.0120
EXP1-MIX	VM	2	2	old	828.13	816.24	32.80	0.0572	0.0748	0.0162
EXP1-MIX	CM	4	0	new	813.60	793.42	30.82	0.0653	0.0633	0.0175
EXP1-MIX	CM	4	1	old	701.76	750.15	31.81	0.0242	0.0366	0.0178
EXP1-MIX	CM	4	2	old	830.48	795.68	38.07	0.0394	0.0547	0.0242
EXP1-MIX	CM	4	3	old	810.21	827.71	33.40	0.0569	0.0852	0.0278
EXP1-MIX	CM	4	4	old	814.83	845.18	31.42	0.1365	0.1071	0.0515
EXP1-MIX	VM	4	0	new	794.71	828.84	27.91	0.0800	0.1239	0.0154
EXP1-MIX	VM	4	1	old	760.95	751.62	43.32	0.0325	0.0367	0.0201
EXP1-MIX	VM	4	2	old	813.88	811.31	35.33	0.0931	0.0684	0.0425
EXP1-MIX	VM	4	3	old	892.21	864.28	35.91	0.1201	0.1357	0.0430
EXP1-MIX	VM	4	4	old	872.57	897.02	33.88	0.1070	0.1910	0.0426
EXP1-MIX	CM	8	0	new	869.11	833.61	28.68	0.1211	0.1407	0.0204
EXP1-MIX	CM	8	1	old	732.04	749.16	45.56	0.0417	0.0365	0.0417
EXP1-MIX	CM	8	2	old	812.09	789.53	32.03	0.0087	0.0499	0.0087
EXP1-MIX	CM	8	3	old	823.56	815.61	41.42	0.0370	0.0708	0.0224
EXP1-MIX	CM	8	4	old	893.50	829.18	41.75	0.1288	0.0852	0.0569
EXP1-MIX	CM	8	5	old	884.32	836.43	44.39	0.1833	0.0937	0.0819
EXP1-MIX	CM	8	6	old	921.84	840.59	44.32	0.1090	0.0989	0.0489
EXP1-MIX	CM	8	7	old	876.15	843.14	46.98	0.2435	0.1022	0.0705

EXP1-MIX	CM	8	8	old	865.45	844.80	32.43	0.1739	0.1044	0.0744
EXP1-MIX	VM	8	0	new	903.23	903.61	25.37	0.2059	0.2659	0.0280
EXP1-MIX	VM	8	1	old	715.69	750.66	47.86	0.0000	0.0366	0.0000
EXP1-MIX	VM	8	2	old	843.37	804.53	42.02	0.0982	0.0610	0.0520
EXP1-MIX	VM	8	3	old	882.48	848.67	52.70	0.1558	0.1101	0.0647
EXP1-MIX	VM	8	4	old	830.33	874.84	36.04	0.1542	0.1491	0.0716
EXP1-MIX	VM	8	5	old	849.65	889.66	35.61	0.1087	0.1740	0.0538
EXP1-MIX	VM	8	6	old	886.96	898.39	44.10	0.1897	0.1895	0.0501
EXP1-MIX	VM	8	7	old	894.73	903.82	36.88	0.1891	0.1995	0.0636
EXP1-MIX	VM	8	8	old	917.74	907.38	42.69	0.2068	0.2062	0.0708
EXP1-VM	VM	2	0	new	709.86	752.04	32.74	0.0850	0.0583	0.0191
EXP1-VM	VM	2	1	old	671.36	716.06	35.55	0.0767	0.0369	0.0304
EXP1-VM	VM	2	2	old	730.79	782.89	32.99	0.0882	0.0781	0.0241
EXP1-VM	VM	4	0	new	809.55	780.98	39.18	0.1133	0.0953	0.0217
EXP1-VM	VM	4	1	old	677.14	715.30	39.01	0.0334	0.0367	0.0193
EXP1-VM	VM	4	2	old	777.29	777.31	39.62	0.0822	0.0707	0.0281
EXP1-VM	VM	4	3	old	835.79	834.61	40.16	0.0880	0.1463	0.0261
EXP1-VM	VM	4	4	old	853.37	870.75	39.83	0.2035	0.2099	0.0580
EXP1-VM	VM	8	0	new	840.68	810.16	32.41	0.2588	0.2019	0.0283
EXP1-VM	VM	8	1	old	755.41	714.38	49.11	0.0869	0.0366	0.0391
EXP1-VM	VM	8	2	old	782.29	770.50	39.32	0.0904	0.0630	0.0384
EXP1-VM	VM	8	3	old	791.50	818.38	37.33	0.1216	0.1186	0.0412
EXP1-VM	VM	8	4	old	825.55	847.39	34.18	0.1771	0.1639	0.0417
EXP1-VM	VM	8	5	old	814.63	863.96	39.10	0.1727	0.1931	0.0460
EXP1-VM	VM	8	6	old	839.76	873.74	39.33	0.3065	0.2114	0.0695
EXP1-VM	VM	8	7	old	870.60	879.84	44.25	0.2264	0.2233	0.0467
EXP1-VM	VM	8	8	old	827.33	883.83	39.65	0.1840	0.2312	0.0492
EXP2-Alternate	AN	2	0	new	777.68	-	65.10	0.0773	-	0.0291
EXP2-Alternate	AN	2	1	old	600.78	-	27.22	0.0260	-	0.0154
EXP2-Alternate	AN	2	2	old	704.43	-	35.77	0.0655	-	0.0204
EXP2-Alternate	CM	2	0	new	753.90	-	57.17	0.0799	-	0.0306
EXP2-Alternate	CM	2	1	old	610.53	-	38.91	0.0278	-	0.0122
EXP2-Alternate	CM	2	2	old	645.54	-	32.63	0.0592	-	0.0167
EXP2-Alternate	AN	4	0	new	745.03	-	45.79	0.0837	-	0.0285
EXP2-Alternate	AN	4	1	old	639.37	-	44.16	0.1061	-	0.0462
EXP2-Alternate	AN	4	2	old	748.09	-	44.88	0.1039	-	0.0380

EXP2-Alternate	AN	4	3	old	858.69	-	55.38	0.0385	-	0.0165
EXP2-Alternate	AN	4	4	old	790.40	-	35.16	0.0803	-	0.0274
EXP2-Alternate	CM	4	0	new	778.91	-	37.62	0.0867	-	0.0239
EXP2-Alternate	CM	4	1	old	664.77	-	41.25	0.0130	-	0.0096
EXP2-Alternate	CM	4	2	old	684.21	-	41.07	0.0199	-	0.0121
EXP2-Alternate	CM	4	3	old	793.07	-	40.56	0.0929	-	0.0307
EXP2-Alternate	CM	4	4	old	767.28	-	39.80	0.0189	-	0.0109
EXP2-Alternate	AN	8	0	new	803.01	-	54.51	0.1128	-	0.0362
EXP2-Alternate	AN	8	1	old	682.97	-	59.03	0.0333	-	0.0186
EXP2-Alternate	AN	8	2	old	712.60	-	43.32	0.0086	-	0.0086
EXP2-Alternate	AN	8	3	old	796.72	-	51.12	0.1530	-	0.0609
EXP2-Alternate	AN	8	4	old	796.00	-	35.21	0.1339	-	0.0497
EXP2-Alternate	AN	8	5	old	883.45	-	64.31	0.0444	-	0.0348
EXP2-Alternate	AN	8	6	old	843.22	-	43.85	0.1756	-	0.0552
EXP2-Alternate	AN	8	7	old	893.20	-	96.24	0.1494	-	0.0551
EXP2-Alternate	AN	8	8	old	873.00	-	94.23	0.1534	-	0.0546
EXP2-Alternate	CM	8	0	new	860.18	-	49.57	0.1306	-	0.0230
EXP2-Alternate	CM	8	1	old	655.43	-	46.46	0.0287	-	0.0204
EXP2-Alternate	CM	8	2	old	803.03	-	71.07	0.0417	-	0.0237
EXP2-Alternate	CM	8	3	old	710.05	-	35.98	0.0172	-	0.0172
EXP2-Alternate	CM	8	4	old	956.02	-	96.08	0.0714	-	0.0323
EXP2-Alternate	CM	8	5	old	757.81	-	46.45	0.1458	-	0.0568
EXP2-Alternate	CM	8	6	old	781.65	-	56.58	0.0923	-	0.0436

EXP2-Alternate	CM	8	7	old	775.64	-	49.18	0.1208	-	0.0484
EXP2-Alternate	CM	8	8	old	733.33	-	43.20	0.0750	-	0.0376
EXP2-ANpure	AN	2	0	new	866.95	694.07	78.43	0.0867	0.0360	0.0245
EXP2-ANpure	AN	2	1	old	711.20	713.49	53.08	0.0327	0.0360	0.0123
EXP2-ANpure	AN	2	2	old	798.32	754.75	63.47	0.0745	0.0457	0.0186
EXP2-ANpure	AN	4	0	new	863.12	712.55	58.73	0.1042	0.0369	0.0242
EXP2-ANpure	AN	4	1	old	726.73	713.09	63.35	0.0468	0.0360	0.0190
EXP2-ANpure	AN	4	2	old	791.76	751.72	51.97	0.0573	0.0439	0.0164
EXP2-ANpure	AN	4	3	old	852.36	795.64	49.96	0.0944	0.0769	0.0258
EXP2-ANpure	AN	4	4	old	929.42	829.79	68.40	0.0872	0.1190	0.0230
EXP2-ANpure	AN	8	0	new	845.89	754.82	43.65	0.1211	0.0496	0.0247
EXP2-ANpure	AN	8	1	old	792.65	712.61	65.34	0.0152	0.0360	0.0112
EXP2-ANpure	AN	8	2	old	846.91	748.02	66.37	0.0305	0.0421	0.0188
EXP2-ANpure	AN	8	3	old	816.36	784.83	53.67	0.0498	0.0655	0.0209
EXP2-ANpure	AN	8	4	old	953.27	811.51	75.98	0.1521	0.0934	0.0493
EXP2-ANpure	AN	8	5	old	925.56	828.72	58.05	0.1089	0.1156	0.0277
EXP2-ANpure	AN	8	6	old	904.44	839.67	64.65	0.1250	0.1314	0.0359
EXP2-ANpure	AN	8	7	old	1003.55	846.82	75.70	0.1617	0.1423	0.0354
EXP2-ANpure	AN	8	8	old	898.03	851.66	56.29	0.1423	0.1499	0.0332
EXP2-CMpure	CM	2	0	new	780.31	665.04	57.71	0.0456	0.0417	0.0152
EXP2-CMpure	CM	2	1	old	663.00	657.16	38.28	0.0407	0.0366	0.0161
EXP2-CMpure	CM	2	2	old	721.99	700.06	44.25	0.0653	0.0535	0.0177
EXP2-CMpure	CM	4	0	new	781.01	694.17	42.91	0.0483	0.0581	0.0127

EXP2-CMpure	CM	4	1	old	637.96	656.36	45.67	0.0276	0.0365	0.0124
EXP2-CMpure	CM	4	2	old	754.00	695.59	42.39	0.0586	0.0499	0.0169
EXP2-CMpure	CM	4	3	old	787.41	720.07	45.90	0.1158	0.0695	0.0268
EXP2-CMpure	CM	4	4	old	829.69	732.58	49.27	0.1201	0.0826	0.0259
EXP2-CMpure	CM	8	0	new	802.87	736.60	37.62	0.1128	0.1300	0.0221
EXP2-CMpure	CM	8	1	old	662.61	655.36	47.06	0.0425	0.0364	0.0232
EXP2-CMpure	CM	8	2	old	749.79	690.21	40.87	0.0379	0.0463	0.0184
EXP2-CMpure	CM	8	3	old	791.62	710.37	65.31	0.0912	0.0598	0.0282
EXP2-CMpure	CM	8	4	old	901.50	720.27	91.41	0.1374	0.0685	0.0332
EXP2-CMpure	CM	8	5	old	831.06	725.41	63.67	0.1684	0.0735	0.0466
EXP2-CMpure	CM	8	6	old	902.32	728.32	75.18	0.1717	0.0765	0.0385
EXP2-CMpure	CM	8	7	old	898.82	730.09	74.68	0.1483	0.0784	0.0357
EXP2-CMpure	CM	8	8	old	846.03	731.24	50.55	0.1542	0.0796	0.0369
EXP2-MIX4	AN	2	0	new	946.04	751.49	61.61	0.0347	0.0360	0.0133
EXP2-MIX4	AN	2	1	old	791.66	777.56	47.56	0.0408	0.0360	0.0175
EXP2-MIX4	AN	2	2	old	842.48	820.99	91.73	0.0796	0.0468	0.0315
EXP2-MIX4	CM	2	0	new	928.85	811.87	65.15	0.0570	0.0491	0.0149
EXP2-MIX4	CM	2	1	old	903.84	788.55	88.33	0.0397	0.0368	0.0154
EXP2-MIX4	CM	2	2	old	949.12	856.74	54.66	0.1059	0.0789	0.0285
EXP2-MIX4	AN	4	0	new	872.91	764.55	43.17	0.0592	0.0362	0.0175
EXP2-MIX4	AN	4	1	old	876.79	777.16	89.51	0.0424	0.0360	0.0215
EXP2-MIX4	AN	4	2	old	896.03	817.49	48.48	0.1059	0.0447	0.0380
EXP2-MIX4	AN	4	3	old	970.60	866.11	73.54	0.1808	0.0838	0.0481
EXP2-MIX4	AN	4	4	old	1080.11	905.72	87.12	0.0845	0.1365	0.0277
EXP2-MIX4	CM	4	0	new	937.71	840.33	43.02	0.1377	0.0742	0.0291
EXP2-MIX4	CM	4	1	old	761.02	787.74	48.40	0.0636	0.0367	0.0353
EXP2-MIX4	CM	4	2	old	956.82	846.77	67.52	0.0692	0.0673	0.0246
EXP2-MIX4	CM	4	3	old	1025.32	899.25	64.94	0.1549	0.1331	0.0377
EXP2-MIX4	CM	4	4	old	1094.68	931.77	74.50	0.1572	0.1875	0.0384
EXP2-MIX4	AN	8	0	new	963.44	792.81	47.75	0.0561	0.0393	0.0199
EXP2-MIX4	AN	8	1	old	1052.89	776.71	88.46	0.0000	0.0360	0.0000
EXP2-MIX4	AN	8	2	old	940.96	813.79	45.25	0.0690	0.0428	0.0410

EXP2-MIX4	AN	8	3	old	1084.43	854.66	85.73	0.0716	0.0708	0.0318
EXP2-MIX4	AN	8	4	old	1188.07	885.65	103.18	0.1672	0.1061	0.0519
EXP2-MIX4	AN	8	5	old	994.06	906.15	63.72	0.2530	0.1350	0.0669
EXP2-MIX4	AN	8	6	old	985.26	919.38	50.01	0.1136	0.1559	0.0360
EXP2-MIX4	AN	8	7	old	1085.85	928.07	57.89	0.0594	0.1704	0.0264
EXP2-MIX4	AN	8	8	old	1024.47	933.95	74.66	0.1544	0.1807	0.0534
EXP2-MIX4	CM	8	0	new	1043.87	875.62	51.25	0.2225	0.1578	0.0307
EXP2-MIX4	CM	8	1	old	878.28	786.78	79.85	0.0287	0.0366	0.0204
EXP2-MIX4	CM	8	2	old	971.16	838.96	81.74	0.0833	0.0593	0.0438
EXP2-MIX4	CM	8	3	old	993.11	880.79	75.02	0.1594	0.1041	0.0512
EXP2-MIX4	CM	8	4	old	996.73	905.31	69.97	0.0931	0.1394	0.0387
EXP2-MIX4	CM	8	5	old	995.15	919.14	94.48	0.0934	0.1618	0.0461
EXP2-MIX4	CM	8	6	old	1132.30	927.28	105.92	0.2051	0.1758	0.0662
EXP2-MIX4	CM	8	7	old	964.72	932.35	70.19	0.1919	0.1847	0.0531
EXP2-MIX4	CM	8	8	old	1092.33	935.67	112.04	0.2024	0.1907	0.0688
EXP2-MIX8	AN	2	0	new	797.53	680.94	73.81	0.0394	0.0360	0.0136
EXP2-MIX8	AN	2	1	old	634.52	707.01	34.43	0.0423	0.0360	0.0139
EXP2-MIX8	AN	2	2	old	735.55	750.45	37.96	0.0392	0.0468	0.0133
EXP2-MIX8	CM	2	0	new	783.30	729.99	52.96	0.0651	0.0427	0.0204
EXP2-MIX8	CM	2	1	old	684.63	712.50	38.88	0.0623	0.0363	0.0232
EXP2-MIX8	CM	2	2	old	782.78	768.48	46.07	0.0462	0.0602	0.0154
EXP2-MIX8	AN	4	0	new	788.78	694.01	52.04	0.0549	0.0362	0.0162
EXP2-MIX8	AN	4	1	old	682.52	706.62	40.45	0.0453	0.0360	0.0323
EXP2-MIX8	AN	4	2	old	919.90	746.94	87.06	0.0382	0.0447	0.0163
EXP2-MIX8	AN	4	3	old	871.05	795.56	43.74	0.0907	0.0838	0.0369
EXP2-MIX8	AN	4	4	old	849.00	835.18	44.31	0.1589	0.1365	0.0469
EXP2-MIX8	CM	4	0	new	839.95	764.56	30.23	0.1213	0.0655	0.0248
EXP2-MIX8	CM	4	1	old	808.98	711.84	59.94	0.0583	0.0362	0.0257
EXP2-MIX8	CM	4	2	old	811.19	759.98	44.39	0.1108	0.0529	0.0431
EXP2-MIX8	CM	4	3	old	916.48	806.33	60.05	0.1542	0.0990	0.0433
EXP2-MIX8	CM	4	4	old	907.45	837.36	73.72	0.1417	0.1434	0.0477
EXP2-MIX8	AN	8	0	new	817.72	722.27	43.56	0.0612	0.0393	0.0115
EXP2-MIX8	AN	8	1	old	801.44	706.16	107.19	0.0357	0.0360	0.0357
EXP2-MIX8	AN	8	2	old	752.41	743.25	49.86	0.0000	0.0428	0.0000
EXP2-MIX8	AN	8	3	old	879.31	784.12	56.31	0.0701	0.0708	0.0327
EXP2-MIX8	AN	8	4	old	987.04	815.11	100.90	0.1580	0.1061	0.0555
EXP2-MIX8	AN	8	5	old	813.33	835.61	32.86	0.1321	0.1350	0.0500
EXP2-MIX8	AN	8	6	old	941.02	848.84	78.93	0.1731	0.1559	0.0623
EXP2-MIX8	AN	8	7	old	1012.52	857.52	93.44	0.1875	0.1704	0.0571
EXP2-MIX8	AN	8	8	old	960.36	863.41	55.79	0.0917	0.1807	0.0424
EXP2-MIX8	CM	8	0	new	905.67	809.93	36.09	0.2076	0.1699	0.0262
EXP2-MIX8	CM	8	1	old	807.98	711.08	84.44	0.0167	0.0362	0.0167
EXP2-MIX8	CM	8	2	old	833.72	753.41	55.73	0.0457	0.0482	0.0220

EXP2-MIX8	CM	8	3	old	868.02	789.60	90.96	0.0565	0.0778	0.0250
EXP2-MIX8	CM	8	4	old	791.78	812.03	36.33	0.0722	0.1044	0.0282
EXP2-MIX8	CM	8	5	old	875.93	825.16	73.93	0.1149	0.1224	0.0524
EXP2-MIX8	CM	8	6	old	890.83	833.06	52.49	0.1333	0.1341	0.0585
EXP2-MIX8	CM	8	7	old	832.36	838.04	60.46	0.1802	0.1417	0.0559
EXP2-MIX8	CM	8	8	old	972.41	841.34	83.00	0.2190	0.1469	0.0596
EXP2- VMpure	VM	2	0	new	850.74	758.36	64.69	0.0556	0.0583	0.0166
EXP2- VMpure	VM	2	1	old	731.10	722.37	40.54	0.0374	0.0369	0.0095
EXP2- VMpure	VM	2	2	old	821.23	789.20	49.39	0.0899	0.0781	0.0188
EXP2- VMpure	VM	4	0	new	825.17	787.29	37.13	0.0873	0.0953	0.0182
EXP2- VMpure	VM	4	1	old	750.93	721.61	36.99	0.0613	0.0367	0.0181
EXP2- VMpure	VM	4	2	old	862.39	783.62	45.06	0.1004	0.0707	0.0233
EXP2- VMpure	VM	4	3	old	957.97	840.92	52.35	0.2307	0.1463	0.0348
EXP2- VMpure	VM	4	4	old	997.06	877.06	71.56	0.1731	0.2099	0.0263
EXP2- VMpure	VM	8	0	new	939.74	816.47	36.36	0.2165	0.2019	0.0215
EXP2- VMpure	VM	8	1	old	771.64	720.69	57.47	0.0490	0.0366	0.0327
EXP2- VMpure	VM	8	2	old	892.84	776.81	45.81	0.1300	0.0630	0.0435
EXP2- VMpure	VM	8	3	old	959.46	824.69	56.98	0.1566	0.1186	0.0389
EXP2- VMpure	VM	8	4	old	907.31	853.70	50.13	0.2007	0.1639	0.0400
EXP2- VMpure	VM	8	5	old	957.41	870.27	50.23	0.2145	0.1931	0.0469
EXP2- VMpure	VM	8	6	old	1074.32	880.05	80.51	0.3416	0.2114	0.0457
EXP2- VMpure	VM	8	7	old	947.05	886.15	53.52	0.3276	0.2233	0.0563
EXP2- VMpure	VM	8	8	old	1091.88	890.14	76.33	0.3181	0.2312	0.0533
PUB-ANpure	AN	3	0	new	752.73	779.07	34.95	0.0499	0.0363	0.0093
PUB-ANpure	AN	3	1	old	685.97	788.80	47.20	0.0110	0.0360	0.0063
PUB-ANpure	AN	3	2	old	813.40	828.54	46.99	0.0421	0.0446	0.0132
PUB-ANpure	AN	3	3	old	852.65	875.07	47.41	0.0320	0.0815	0.0114
PUB-ANpure	AN	6	0	new	813.82	807.57	38.65	0.0766	0.0402	0.0133
PUB-ANpure	AN	6	1	old	662.77	788.36	46.12	0.0000	0.0360	0.0000

PUB-ANpure	AN	6	2	old	804.80	825.24	59.12	0.0222	0.0429	0.0222
PUB-ANpure	AN	6	3	old	895.27	865.20	60.90	0.0678	0.0703	0.0271
PUB-ANpure	AN	6	4	old	983.40	895.12	59.56	0.1017	0.1041	0.0383
PUB-ANpure	AN	6	5	old	926.25	914.75	55.67	0.1321	0.1315	0.0345
PUB-ANpure	AN	6	6	old	877.36	927.38	48.65	0.1374	0.1511	0.0398
PUB-ANpure	AN	9	0	new	836.73	842.87	39.19	0.0875	0.0585	0.0115
PUB-ANpure	AN	9	1	old	743.79	788.05	52.58	0.0119	0.0360	0.0119
PUB-ANpure	AN	9	2	old	829.29	822.79	42.29	0.0201	0.0418	0.0141
PUB-ANpure	AN	9	3	old	931.60	858.21	60.48	0.0230	0.0635	0.0160
PUB-ANpure	AN	9	4	old	867.17	883.51	57.02	0.1563	0.0890	0.0602
PUB-ANpure	AN	9	5	old	976.78	899.68	55.26	0.1367	0.1091	0.0542
PUB-ANpure	AN	9	6	old	900.83	909.94	53.90	0.1862	0.1233	0.0588
PUB-ANpure	AN	9	7	old	989.60	916.62	55.25	0.1977	0.1331	0.0611
PUB-ANpure	AN	9	8	old	960.95	921.12	62.46	0.1586	0.1400	0.0484
PUB-ANpure	AN	9	9	old	947.67	924.27	56.88	0.1039	0.1449	0.0423
PUB-CMpure	CM	3	0	new	545.12	579.76	17.91	0.0024	0.0362	0.0017
PUB-CMpure	CM	3	1	old	564.23	601.21	35.56	0.0042	0.0362	0.0042
PUB-CMpure	CM	3	2	old	648.47	628.10	36.82	0.0250	0.0415	0.0115
PUB-CMpure	CM	3	3	old	644.22	642.33	40.36	0.0306	0.0482	0.0134
PUB-CMpure	CM	6	0	new	550.02	595.79	19.96	0.0144	0.0373	0.0061
PUB-CMpure	CM	6	1	old	546.32	600.55	36.51	0.0056	0.0362	0.0056
PUB-CMpure	CM	6	2	old	611.69	625.18	47.93	0.0000	0.0403	0.0000
PUB-CMpure	CM	6	3	old	676.47	637.59	47.22	0.0222	0.0453	0.0134
PUB-CMpure	CM	6	4	old	681.97	643.30	40.52	0.0083	0.0484	0.0083
PUB-CMpure	CM	6	5	old	671.37	646.18	41.66	0.0243	0.0501	0.0130
PUB-CMpure	CM	6	6	old	675.17	647.78	44.80	0.0469	0.0511	0.0182
PUB-CMpure	CM	9	0	new	570.10	613.63	23.06	0.0040	0.0410	0.0023
PUB-CMpure	CM	9	1	old	600.78	600.04	53.18	0.0000	0.0361	0.0000
PUB-CMpure	CM	9	2	old	590.28	623.01	36.04	0.0000	0.0395	0.0000
PUB-CMpure	CM	9	3	old	645.09	634.14	44.45	0.0172	0.0435	0.0127
PUB-CMpure	CM	9	4	old	642.41	639.17	44.99	0.0536	0.0459	0.0393
PUB-CMpure	CM	9	5	old	615.97	641.68	34.52	0.0057	0.0472	0.0057
PUB-CMpure	CM	9	6	old	656.19	643.07	47.53	0.0316	0.0480	0.0177
PUB-CMpure	CM	9	7	old	630.04	643.91	44.73	0.0660	0.0484	0.0395
PUB-CMpure	CM	9	8	old	734.77	644.45	57.10	0.0864	0.0488	0.0428
PUB-CMpure	CM	9	9	old	664.61	644.82	50.25	0.0476	0.0490	0.0372
PUB-MIX1	AN	3	0	new	695.58	734.48	20.38	0.0182	0.0368	0.0058
PUB-MIX1	AN	3	1	old	653.81	736.81	24.99	0.0300	0.0360	0.0107
PUB-MIX1	AN	3	2	old	715.27	775.45	25.15	0.0334	0.0441	0.0100
PUB-MIX1	AN	3	3	old	856.45	818.94	33.59	0.0912	0.0771	0.0212
PUB-MIX1	CM	3	0	new	639.37	726.87	15.18	0.0144	0.0365	0.0057
PUB-MIX1	CM	3	1	old	682.12	743.16	28.92	0.0101	0.0363	0.0064
PUB-MIX1	CM	3	2	old	801.15	783.58	27.07	0.0870	0.0491	0.0200

PUB-MIX1	CM	3	3	old	855.44	812.91	30.98	0.1341	0.0730	0.0235
PUB-MIX1	VM	3	0	new	798.85	812.86	26.23	0.1000	0.0947	0.0144
PUB-MIX1	VM	3	1	old	690.72	746.23	29.04	0.0332	0.0366	0.0118
PUB-MIX1	VM	3	2	old	798.35	803.60	27.39	0.0948	0.0650	0.0202
PUB-MIX1	VM	3	3	old	837.52	856.12	33.71	0.1655	0.1297	0.0293
PUB-MIX1	AN	6	0	new	728.87	770.72	20.88	0.0374	0.0457	0.0099
PUB-MIX1	AN	6	1	old	700.78	736.37	33.29	0.0199	0.0360	0.0109
PUB-MIX1	AN	6	2	old	760.85	772.22	32.19	0.0689	0.0425	0.0199
PUB-MIX1	AN	6	3	old	831.93	809.70	33.64	0.1218	0.0670	0.0268
PUB-MIX1	AN	6	4	old	857.13	836.98	33.12	0.1255	0.0963	0.0303
PUB-MIX1	AN	6	5	old	880.50	854.61	32.81	0.0932	0.1197	0.0262
PUB-MIX1	AN	6	6	old	897.08	865.85	38.02	0.1233	0.1362	0.0317
PUB-MIX1	CM	6	0	new	662.48	743.87	16.86	0.0240	0.0384	0.0074
PUB-MIX1	CM	6	1	old	734.73	742.48	32.98	0.0254	0.0363	0.0146
PUB-MIX1	CM	6	2	old	793.38	778.94	33.37	0.0437	0.0460	0.0170
PUB-MIX1	CM	6	3	old	862.81	803.35	39.77	0.1088	0.0629	0.0255
PUB-MIX1	CM	6	4	old	816.36	816.35	36.47	0.1442	0.0753	0.0342
PUB-MIX1	CM	6	5	old	898.35	823.38	40.13	0.1982	0.0828	0.0381
PUB-MIX1	CM	6	6	old	833.88	827.45	31.38	0.1749	0.0875	0.0336
PUB-MIX1	VM	6	0	new	899.66	841.89	25.55	0.2051	0.2001	0.0160
PUB-MIX1	VM	6	1	old	728.09	745.43	38.42	0.0071	0.0365	0.0052
PUB-MIX1	VM	6	2	old	778.68	798.06	31.32	0.1115	0.0591	0.0307
PUB-MIX1	VM	6	3	old	868.24	843.11	35.79	0.1227	0.1084	0.0311
PUB-MIX1	VM	6	4	old	916.17	870.65	39.15	0.1214	0.1494	0.0279
PUB-MIX1	VM	6	5	old	864.01	886.52	33.34	0.1967	0.1761	0.0361
PUB-MIX1	VM	6	6	old	882.27	895.94	36.71	0.1641	0.1930	0.0359
PUB-MIX1	AN	9	0	new	749.96	812.61	18.98	0.0250	0.0837	0.0075
PUB-MIX1	AN	9	1	old	694.17	736.05	36.71	0.0311	0.0360	0.0195
PUB-MIX1	AN	9	2	old	825.89	769.74	36.04	0.0238	0.0414	0.0176
PUB-MIX1	AN	9	3	old	832.31	802.86	36.96	0.1289	0.0607	0.0369
PUB-MIX1	AN	9	4	old	850.45	825.87	32.81	0.0887	0.0826	0.0265
PUB-MIX1	AN	9	5	old	883.58	840.34	38.67	0.1423	0.0996	0.0374
PUB-MIX1	AN	9	6	old	940.94	849.44	42.45	0.1582	0.1114	0.0408
PUB-MIX1	AN	9	7	old	913.59	855.34	41.90	0.1680	0.1196	0.0448
PUB-MIX1	AN	9	8	old	935.06	859.31	48.29	0.1213	0.1252	0.0352
PUB-MIX1	AN	9	9	old	892.21	862.07	43.76	0.1664	0.1293	0.0399
PUB-MIX1	CM	9	0	new	670.84	762.19	16.12	0.0263	0.0435	0.0084
PUB-MIX1	CM	9	1	old	689.26	741.98	37.09	0.0338	0.0362	0.0200
PUB-MIX1	CM	9	2	old	761.70	776.00	32.34	0.0321	0.0444	0.0181
PUB-MIX1	CM	9	3	old	850.84	797.69	37.09	0.0572	0.0578	0.0223
PUB-MIX1	CM	9	4	old	761.76	808.93	31.41	0.0582	0.0673	0.0227
PUB-MIX1	CM	9	5	old	797.04	814.93	29.19	0.0997	0.0730	0.0346
PUB-MIX1	CM	9	6	old	888.90	818.37	45.49	0.1211	0.0766	0.0340

PUB-MIX1	CM	9	7	old	817.71	820.48	37.80	0.0853	0.0788	0.0309
PUB-MIX1	CM	9	8	old	866.43	821.85	43.01	0.1816	0.0802	0.0426
PUB-MIX1	CM	9	9	old	859.44	822.79	42.87	0.1264	0.0813	0.0376
PUB-MIX1	VM	9	0	new	975.96	844.82	27.87	0.3402	0.3297	0.0192
PUB-MIX1	VM	9	1	old	688.68	744.82	36.75	0.0351	0.0364	0.0246
PUB-MIX1	VM	9	2	old	830.06	793.76	37.65	0.0054	0.0553	0.0054
PUB-MIX1	VM	9	3	old	817.91	833.27	40.24	0.0643	0.0943	0.0253
PUB-MIX1	VM	9	4	old	852.25	856.65	41.34	0.1714	0.1259	0.0414
PUB-MIX1	VM	9	5	old	901.38	869.92	42.89	0.1079	0.1462	0.0308
PUB-MIX1	VM	9	6	old	935.43	877.77	45.73	0.1231	0.1590	0.0375
PUB-MIX1	VM	9	7	old	911.90	882.67	46.45	0.2299	0.1673	0.0467
PUB-MIX1	VM	9	8	old	869.25	885.89	47.86	0.2083	0.1728	0.0482
PUB-MIX1	VM	9	9	old	904.19	888.10	41.63	0.1538	0.1766	0.0366
PUB-MIX2	AN	3	0	new	780.61	845.28	29.32	0.0164	0.0368	0.0070
PUB-MIX2	AN	3	1	old	762.01	847.61	33.69	0.0212	0.0360	0.0072
PUB-MIX2	AN	3	2	old	862.13	886.24	34.04	0.0627	0.0441	0.0177
PUB-MIX2	AN	3	3	old	913.56	929.74	34.32	0.0802	0.0771	0.0203
PUB-MIX2	CM	3	0	new	729.46	835.25	24.47	0.0138	0.0363	0.0046
PUB-MIX2	CM	3	1	old	843.73	853.81	33.73	0.0165	0.0363	0.0094
PUB-MIX2	CM	3	2	old	924.76	893.24	38.05	0.1045	0.0484	0.0279
PUB-MIX2	CM	3	3	old	906.37	921.27	36.35	0.1139	0.0705	0.0254
PUB-MIX2	VM	3	0	new	940.60	923.66	32.04	0.1200	0.0947	0.0195
PUB-MIX2	VM	3	1	old	828.29	857.03	34.14	0.0061	0.0366	0.0043
PUB-MIX2	VM	3	2	old	910.73	914.39	33.26	0.0624	0.0650	0.0202
PUB-MIX2	VM	3	3	old	949.00	966.92	38.25	0.1502	0.1297	0.0285
PUB-MIX2	AN	6	0	new	820.39	881.51	28.41	0.0169	0.0457	0.0056
PUB-MIX2	AN	6	1	old	785.79	847.16	35.78	0.0110	0.0360	0.0082
PUB-MIX2	AN	6	2	old	896.32	883.01	38.92	0.0781	0.0425	0.0315
PUB-MIX2	AN	6	3	old	891.12	920.49	34.76	0.0435	0.0670	0.0217
PUB-MIX2	AN	6	4	old	955.15	947.78	34.36	0.0594	0.0963	0.0228
PUB-MIX2	AN	6	5	old	994.06	965.41	39.77	0.1690	0.1197	0.0402
PUB-MIX2	AN	6	6	old	1041.24	976.65	43.01	0.0918	0.1362	0.0324
PUB-MIX2	CM	6	0	new	779.69	850.98	28.62	0.0186	0.0378	0.0062
PUB-MIX2	CM	6	1	old	813.97	853.14	46.15	0.0000	0.0363	0.0000
PUB-MIX2	CM	6	2	old	867.07	888.75	39.12	0.0938	0.0455	0.0350
PUB-MIX2	CM	6	3	old	949.97	912.16	37.00	0.0487	0.0612	0.0209
PUB-MIX2	CM	6	4	old	932.93	924.51	48.71	0.1260	0.0726	0.0355
PUB-MIX2	CM	6	5	old	981.09	931.16	46.54	0.1895	0.0795	0.0415
PUB-MIX2	CM	6	6	old	976.71	934.99	39.04	0.1269	0.0837	0.0331
PUB-MIX2	VM	6	0	new	1064.01	952.68	33.55	0.2813	0.2001	0.0226
PUB-MIX2	VM	6	1	old	850.52	856.22	36.58	0.0178	0.0365	0.0103
PUB-MIX2	VM	6	2	old	899.07	908.86	42.24	0.0471	0.0591	0.0163
PUB-MIX2	VM	6	3	old	906.07	953.91	39.34	0.1497	0.1084	0.0396

PUB-MIX2	VM	6	4	old	1012.69	981.45	42.57	0.1282	0.1494	0.0332
PUB-MIX2	VM	6	5	old	913.81	997.31	38.68	0.2016	0.1761	0.0438
PUB-MIX2	VM	6	6	old	1005.69	1006.74	46.29	0.1641	0.1930	0.0354
PUB-MIX2	AN	9	0	new	847.48	923.40	26.16	0.0262	0.0837	0.0070
PUB-MIX2	AN	9	1	old	879.67	846.84	51.45	0.0000	0.0360	0.0000
PUB-MIX2	AN	9	2	old	851.67	880.53	42.88	0.0625	0.0414	0.0320
PUB-MIX2	AN	9	3	old	1033.88	913.65	46.71	0.1000	0.0607	0.0379
PUB-MIX2	AN	9	4	old	977.32	936.67	49.58	0.0943	0.0826	0.0420
PUB-MIX2	AN	9	5	old	1067.91	951.14	53.13	0.1758	0.0996	0.0461
PUB-MIX2	AN	9	6	old	1006.80	960.24	50.17	0.1635	0.1114	0.0509
PUB-MIX2	AN	9	7	old	1018.52	966.14	42.08	0.0861	0.1196	0.0292
PUB-MIX2	AN	9	8	old	1095.63	970.10	53.34	0.1607	0.1252	0.0499
PUB-MIX2	AN	9	9	old	1064.38	972.87	47.91	0.1408	0.1293	0.0428
PUB-MIX2	CM	9	0	new	806.82	867.91	28.67	0.0287	0.0416	0.0076
PUB-MIX2	CM	9	1	old	812.82	852.65	48.98	0.0000	0.0362	0.0000
PUB-MIX2	CM	9	2	old	883.23	885.91	42.73	0.0000	0.0439	0.0000
PUB-MIX2	CM	9	3	old	953.67	906.75	41.18	0.0566	0.0564	0.0258
PUB-MIX2	CM	9	4	old	929.71	917.46	48.49	0.0882	0.0652	0.0363
PUB-MIX2	CM	9	5	old	921.82	923.15	40.45	0.0879	0.0705	0.0364
PUB-MIX2	CM	9	6	old	929.95	926.41	47.38	0.0932	0.0737	0.0352
PUB-MIX2	CM	9	7	old	928.61	928.40	50.43	0.1374	0.0757	0.0426
PUB-MIX2	CM	9	8	old	928.76	929.70	56.03	0.1590	0.0771	0.0466
PUB-MIX2	CM	9	9	old	968.96	930.58	52.82	0.1176	0.0780	0.0386
PUB-MIX2	VM	9	0	new	1084.85	955.62	35.52	0.4122	0.3297	0.0247
PUB-MIX2	VM	9	1	old	893.21	855.62	57.92	0.0064	0.0364	0.0064
PUB-MIX2	VM	9	2	old	931.42	904.55	43.41	0.0389	0.0553	0.0207
PUB-MIX2	VM	9	3	old	901.84	944.07	38.85	0.1071	0.0943	0.0397
PUB-MIX2	VM	9	4	old	936.91	967.44	47.89	0.0784	0.1259	0.0344
PUB-MIX2	VM	9	5	old	954.60	980.72	48.08	0.0833	0.1462	0.0314
PUB-MIX2	VM	9	6	old	996.30	988.57	54.07	0.2255	0.1590	0.0566
PUB-MIX2	VM	9	7	old	926.66	993.47	46.09	0.1146	0.1673	0.0391
PUB-MIX2	VM	9	8	old	1030.80	996.69	57.14	0.2138	0.1728	0.0528
PUB-MIX2	VM	9	9	old	1023.03	998.89	55.16	0.1384	0.1766	0.0432
PUB-VMpure	VM	3	0	new	989.25	1056.20	52.62	0.0410	0.0700	0.0134
PUB-VMpure	VM	3	1	old	921.50	1005.44	59.36	0.0114	0.0366	0.0088
PUB-VMpure	VM	3	2	old	1056.50	1065.69	55.80	0.0539	0.0678	0.0200
PUB-VMpure	VM	3	3	old	1035.63	1124.00	56.52	0.0366	0.1432	0.0132
PUB-VMpure	VM	6	0	new	1095.62	1089.87	51.71	0.1205	0.1371	0.0199

PUB- VMpure	VM	6	1	old	923.23	1004.67	67.24	0.0000	0.0365	0.0000
PUB- VMpure	VM	6	2	old	1127.48	1059.93	62.74	0.0475	0.0614	0.0242
PUB- VMpure	VM	6	3	old	1151.86	1109.71	73.54	0.1185	0.1188	0.0376
PUB- VMpure	VM	6	4	old	1178.59	1141.05	64.96	0.1573	0.1681	0.0425
PUB- VMpure	VM	6	5	old	1162.48	1159.29	66.87	0.1842	0.2008	0.0522
PUB- VMpure	VM	6	6	old	1191.85	1170.16	70.39	0.2006	0.2216	0.0515
PUB- VMpure	VM	9	0	new	1125.28	1105.27	52.95	0.1915	0.2339	0.0242
PUB- VMpure	VM	9	1	old	949.39	1004.10	62.90	0.0093	0.0364	0.0093
PUB- VMpure	VM	9	2	old	1020.96	1055.66	57.83	0.0119	0.0573	0.0119
PUB- VMpure	VM	9	3	old	1066.12	1099.55	50.25	0.0929	0.1034	0.0449
PUB- VMpure	VM	9	4	old	1162.61	1126.30	56.57	0.1177	0.1419	0.0369
PUB- VMpure	VM	9	5	old	1131.05	1141.70	62.00	0.1856	0.1671	0.0557
PUB- VMpure	VM	9	6	old	1179.52	1150.86	63.05	0.1339	0.1830	0.0474
PUB- VMpure	VM	9	7	old	1237.50	1156.58	61.68	0.3093	0.1934	0.0667
PUB- VMpure	VM	9	8	old	1101.04	1160.35	56.88	0.1494	0.2004	0.0484
PUB- VMpure	VM	9	9	old	1204.87	1162.93	64.22	0.1667	0.2052	0.0505